Complete information orders of quantum channels: equivalence and undecidability*

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Abstract

With arbitrary finite quantum references, relative entropy, Holevo information and SLD Fisher information induce the same channel order, reversed by complementation. This order is coRE-complete, even for strictly positive rational POVMs. Its bounded entropy defect is additive and uncomputable at every fixed positive additive accuracy.

1 Introduction

When is one quantum channel more informative than another? A physical answer asks whether the second output can be obtained by processing the first. A statistical answer asks whether the first preserves every distinction at least as well. The latter is less demanding, but apparently ambiguous: one may compare two states by relative entropy, an ensemble by its Holevo information, or a small perturbation by Fisher information.

We show that these statistical answers coincide when the comparison allows every finite untouched quantum reference. In particular, the global relative-entropy order is exactly the local symmetric logarithmic derivative (SLD) order. This is not an ordinary passage from a function to its Hessian: the entropy Hessian is the BKM metric, which differs from SLD on noncommuting perturbations. The reference supplies the missing freedom. Coherent blocks turn local SLD tests into resolvents of two states; conversely, a reference chosen at an equality point of the entropy comparison isolates the SLD kernel from a denominator Hessian.

The local description reveals a second structure. Taking Stinespring complements reverses the order: the receiver that preserves more information has an environment that preserves less. The reason is a purification identity. For each input perturbation there is a dual phase experiment, the same for both channels, whose environmental Fisher information is exactly the receiver's loss. Complete comparison lets us use that dual experiment as another admissible test.

Neither characterization makes comparison computable. We prove coRE-completeness, already for measurements with strictly positive rational-complex effects. Every failed comparison has a finite witness, but no computable function of the measurement pair bounds a sufficient SLD reference dimension. More strongly, even the bounded entropy advantage of one measurement over another cannot be computed to error $1/4$, or to any other fixed positive rational error. Exact additivity amplifies a small, effectively controlled witness into this fixed-error obstruction. Theorems 2.2 and 2.3 state the two computability results precisely.

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The reduction is dictated by the geometry of the SLD test. A measurement output contributes an inverse Sylvester operator, so its variables are not arbitrary noncommuting matrices: they share two positive baselines and a spatial tensor-product structure. We first recover two local players inside this constrained domain. Opposite Pauli representations pin down a balanced Bell sector, and a local averaging argument turns approximate observables into genuine game measurements. A multiplication graph then expresses the game polynomial using only inverse Sylvester kernels. Finally, paired measurement outcomes and additional local measurements extend the comparison from a selected tangent subspace to every input and tangent. Figure 1 follows this chain. The construction never needs the strategy or its dimension.

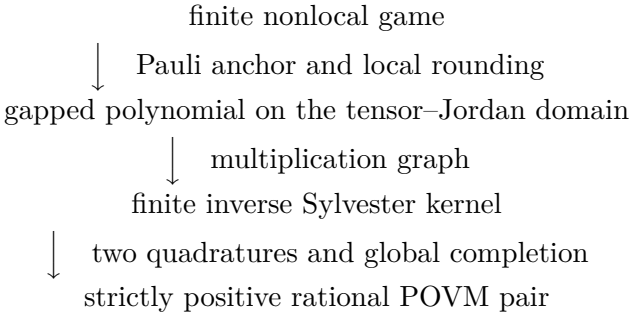


Figure 1: The game determines the measurement pair, without a bound on the strategy dimension. Low game value gives complete dominance; a high-scoring finite strategy gives a strict violation.

Relation to earlier work. Quantum less-noisy and more-capable comparisons were developed by Watanabe [4]. Hirche, Rouzé and Stilck França [2] identify relative-entropy and Holevo comparisons and prove complete tensorization; Hirche and Leditzky [5] develop their relation to complementary channels and capacities. The distinction between the SLD and BKM metrics belongs to the theory of monotone quantum metrics [6, 7]. The SLD metric also describes optimized classical Fisher information and measured chi-squared divergence [8, 9]. Hirche's recent preprint [3] proves the forward complete measured-chi-squared-to-entropy hierarchy. We supply the general reverse implication by a saturated reference and a finite resolvent filter. The complementary-parameter information identity of Faist et al. [10] is the metrological predecessor of our common-dual-input argument.

Our external complexity input is the precise finite perfect-strategy form of $\mathrm{MIP}^* = \mathrm{RE}$ [1]. Its finite attainment statement matters for the quantitative result: entangled value one alone need not be attained [11]. All steps from that theorem to the measurement orders are proved below. The main text develops the constructions and their logical dependence; Appendices A and B contain the longer quantitative estimates.

2 The orders and the results

All logarithms in this paper are natural. Changing their base changes numerical information units but not any order. All Hilbert spaces in individual tests are finite dimensional; there is no uniform bound on the dimensions of references. A state is a positive semidefinite matrix of trace one, and is *faithful* when positive definite. A channel is a completely positive trace-preserving linear map. We use the unnormalized Choi convention $J(\Phi) = \sum_{i,j} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|)$. Matrices

with *rational complex entries* have real and imaginary parts in $\mathbb{Q}$.

For $W > 0$, write

$$\Omega_W(X) = \frac{WX + XW}{2}, \qquad g_W(X) = \langle X, \Omega_W^{-1}(X) \rangle_{\mathrm{HS}}. \tag{1}$$

For Hermitian X, this is the symmetric logarithmic derivative (SLD) quadratic form. In an eigenbasis of W,

$$g_W(X) = \sum_{i,j} \frac{2|X_{ij}|^2}{w_i + w_j}.$$

When W and X are diagonal this is $\sum_i X_{ii}^2/w_i$, the classical Fisher information of the probability perturbation $w_i + tX_{ii}$. Off-diagonal entries describe changes of eigenvectors; their cost is weighted by the mean population of the two levels. We reserve $\mathcal{J}_{U,V}$ for the *unhalved* rectangular Sylvester operator

$$\mathcal{J}_{U,V}(Z) = UZ + ZV. \tag{2}$$

Thus $\Omega_W = \mathcal{J}_{W,W}/2$. This factor of two matters in the POVM construction.

A POVM $E = (E_i)$ on $\mathbb{C}^a$ has $E_i \geq 0$ and $\sum_i E_i = I_a$. Its measurement channel is

$$\mathcal{E}(X) = \sum_i \mathrm{Tr}(E_i X)\, |i\rangle\langle i|.$$

For an untouched reference R, its output on a state ρ has blocks $\rho_i^E = \mathrm{Tr}_a[(E_i \otimes I_R)\rho]$. Strict positivity means *every* $E_i > 0$.

Definition 2.1. For channels $\mathcal{P}, \mathcal{Q}$ with a common input, write $\mathcal{P} \succeq^{\mathrm{c}}_{\mathrm{SLD}} \mathcal{Q}$ when

$$g_{\mathcal{P}_R(\rho)}(\mathcal{P}_R(H)) \geq g_{\mathcal{Q}_R(\rho)}(\mathcal{Q}_R(H)) \tag{3}$$

for every finite reference R, faithful input state ρ, and traceless Hermitian tangent H, where $\mathcal{P}_R = \mathcal{P} \otimes \mathrm{id}_R$. Write $\mathcal{P} \succeq^{\mathrm{c}}_{D} \mathcal{Q}$ when

$$D(\mathcal{P}_R(\alpha) \| \mathcal{P}_R(\beta)) \geq D(\mathcal{Q}_R(\alpha) \| \mathcal{Q}_R(\beta)) \tag{4}$$

for every finite R and every pair of input states α, β. Here $D(A\|B) = \mathrm{Tr}\, A(\log A - \log B)$ when $\mathrm{supp}\, A \subseteq \mathrm{supp}\, B$, and $+\infty$ otherwise, with the usual convention at zero eigenvalues.

The two reference marginals in (4) may differ. This unrestricted convention is used throughout. In (3) we restrict output matrices to the fixed support of the channel's output on the identity, where they are faithful. For strictly positive POVMs the output blocks of faithful inputs are already positive definite.

As a basic example, the identity channel dominates a channel that discards its input, by data processing. This remains true with a reference: discarding the input leaves only the reference marginal. Taking complements exchanges these two channels. Theorem 4.2 extends that elementary reversal to arbitrary channel pairs, even when neither output can physically simulate the other.

Theorem 2.2 (Undecidability)**.** *For finite rational-complex strictly positive POVM pairs (E, F) on a common input, the set satisfying $\mathcal{E} \succeq^{\mathrm{c}}_{\mathrm{SLD}} \mathcal{F}$ is* coRE*-complete under computable many-one reductions, and its complement is* RE*-complete. The input dimension and outcome counts are part of the description. The same classifications hold for unrestricted complete Umegaki order, and for both orders on arbitrary channels given by exact rational-complex Choi matrices.*

There is no total computable function of a valid rational strictly positive POVM pair that bounds a reference dimension sufficient to witness every failure of its complete SLD order. In particular, these orders are undecidable, and their dominating sets are not recursively enumerable.

Every SLD failure has some finite witness. The reference obstruction rules out a *computable sufficient bound* from the input, not finite witnesses themselves.

Here RE denotes the recursively enumerable sets: membership has a search that eventually accepts every member, though it may run forever on a nonmember. A set is in coRE when its complement is in RE. Completeness is under computable many-one reductions. The theorem concerns variable dimensions and generic complex effects; it makes no hardness claim for fixed input dimension or real matrices.

The quantitative result concerns an ordinary finite-valued functional. For strictly positive POVMs define the target's entropy advantage by

$$\mathcal{V}(E,F) = \sup_{R,\rho,\sigma} \left[D(\mathcal{F}_R(\rho)\|\mathcal{F}_R(\sigma)) - D(\mathcal{E}_R(\rho)\|\mathcal{E}_R(\sigma)) \right], \tag{5}$$

where R is any finite reference and ρ, σ are faithful input states. Every difference is between finite numbers. We prove in Section 8.1 that $\mathcal{V}$ is bounded, nonnegative and exactly additive under tensor products, and vanishes precisely on dominating pairs.

Theorem 2.3 (Inapproximability at fixed additive error)**.** *For every fixed rational $c > 0$, there is no total algorithm that, given a finite rational-complex strictly positive POVM pair (E, F), returns a rational v satisfying*

$$|v - \mathcal{V}(E,F)| \le c.$$

In particular this holds for $c = 1/4$.

Thus the obstruction survives a fixed numerical tolerance. The proof needs more than qualitative undecidability: a perfect-strategy sector gives an entropy margin computable without knowing the strategy dimension, and tensor additivity amplifies that margin. These two steps are the purpose of Section 8.

3 Global, local and estimation comparisons

For commuting states, Fisher information is the Hessian of relative entropy, and integration along a line recovers the divergence. In the quantum setting this argument recovers BKM information, not SLD information. The distinction is already visible in their spectral coefficients. Understanding how a reference removes it is the main point of this section.

For $A > 0$, let

$$\mathcal{K}_A(X) = \int_0^1 A^t X A^{1-t}\, dt, \qquad b_A(H) = \langle H, \mathcal{K}_A^{-1}(H) \rangle_{\mathrm{HS}}$$

be the BKM quadratic form. In an eigenbasis of A,

$$b_A(H) = \sum_{i,j} \frac{\log a_i - \log a_j}{a_i - a_j} |H_{ij}|^2,$$

where the coefficient is $1/a_i$ when $a_i = a_j$. The BKM and SLD forms agree for perturbations commuting with the base. On coherences, the logarithmic mean in the BKM denominator replaces the arithmetic mean in SLD. The reference filter below bridges these two means.

A two-level example. Take

$$A = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad H = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The states $A + tH$ are faithful for $|t| < \sqrt{3}/2$. Substituting the two off-diagonal entries into the spectral formulas gives

$$g_A(H) = 4|H_{12}|^2 = 1, \qquad b_A(H) = 4\log 3\,|H_{12}|^2 = \log 3.$$

Accordingly $D(A + tH\|A) = \frac{1}{2}t^2 \log 3 + O(t^4)$: the ordinary entropy expansion retains the larger BKM cost of this coherence. An independent reference state $\eta > 0$ leaves both costs unchanged: $g_{A\otimes\eta}(H \otimes \eta) = 1$ and $b_{A\otimes\eta}(H \otimes \eta) = \log 3$. The correlated reference in (10) supplies a different family of tests. At its unperturbed pair, relative entropy is independent of the channel, while reference coherences give a denominator Hessian with adjustable kernels $(a_j + a_i t)^{-1}$, as in (12). Concentrating their weight at $t = 1$ selects the SLD coefficient $2/(a_i + a_j)$. This extra freedom is what turns an entropy comparison into an SLD comparison despite the numerical mismatch above.

The Holevo information of an ensemble $\mathsf{E} = \{p_x, \rho_x\}$ is $\chi(\mathsf{E}) = \sum_x p_x D(\rho_x\|\bar{\rho})$, where $\bar{\rho} = \sum_x p_x \rho_x$. The measured chi-squared divergence is the supremum of the classical Pearson divergence over finite POVMs.

Theorem 3.1 (Equivalent complete orders)**.** *For any finite-dimensional channels $\mathcal{P}, \mathcal{Q}$ with a common input, the following comparisons are equivalent:*

1. *unrestricted complete Umegaki order $\mathcal{P} \succeq^{\mathrm{c}}_{D} \mathcal{Q}$;*
2. *Holevo information after $\mathcal{P}$ is at least that after $\mathcal{Q}$ for every finite ensemble with every finite untouched reference; equiprobable binary ensembles suffice;*
3. *$b_{\mathcal{P}_R(\rho)}(\mathcal{P}_R(H)) \geq b_{\mathcal{Q}_R(\rho)}(\mathcal{Q}_R(H))$ for every finite reference, faithful state and traceless Hermitian tangent;*
4. *complete SLD order $\mathcal{P} \succeq^{\mathrm{c}}_{\mathrm{SLD}} \mathcal{Q}$;*
5. *measured chi-squared divergence after $\mathcal{P}$ is at least that after $\mathcal{Q}$, with every finite reference and every state pair whose denominator is faithful.*

No equality of the two reference marginals is imposed. These orders tensorize, including for entangled inputs across the factors.

Proof of the standard links and tensorization. The identity defining χ gives entropy-to-Holevo comparison. To recover the local metric, an equiprobable binary ensemble is enough. For a faithful state A and traceless Hermitian H,

$$\chi(\{\tfrac{1}{2}, A + tH; \tfrac{1}{2}, A - tH\}) = \tfrac{t^2}{2} b_A(H) + o(t^2).$$

Both states are faithful for sufficiently small $|t|$. Comparing their Holevo information after the channels and letting $t \to 0$ gives BKM comparison. Conversely, for faithful A, B,

$$D(A\|B) = \int_0^1 (1-t) b_{B+t(A-B)}(A - B)\, dt.$$

Thus BKM comparison recovers entropy comparison. A common faithful regularization recovers boundary state pairs, including a strict violation with infinite larger divergence. These are the familiar Hessian and affine-path formulas for Umegaki divergence [6, 7, 2]. In particular every failure has a strict finite equiprobable binary Holevo witness.

For faithful σ, optimized classical chi-squared equals $g_\sigma(\rho-\sigma)$ [9]. To see why, put $L=\Omega_\sigma^{-1}(\rho-\sigma)$. For a POVM (M_j), Cauchy–Schwarz gives

$$\sum_j \frac{[\operatorname{Tr} M_j(\rho-\sigma)]^2}{\operatorname{Tr} M_j\sigma} \le \operatorname{Tr}\sigma L^2 = g_\sigma(\rho-\sigma),$$

and the spectral measurement of L attains equality. Tangent scaling $\rho=\sigma+tH$ for sufficiently small t proves the equivalence with SLD order. The entropy/SLD link is proved in full below.

For tensorization, replace one channel factor at a time. The other unprocessed inputs and already produced outputs, together with the original reference, form an allowed untouched reference for that replacement. Chaining the one-factor entropy inequalities proves the product comparison [2]. □

It remains to connect entropy and SLD. The two directions use the reference differently. An off-diagonal qubit tangent encodes a resolvent of two states, which integrates to entropy. For the reverse direction, we choose an entropy test that is already an equality for every channel, then perturb its denominator. Nonnegativity forces its Hessian to have the required sign; a finite reference filter selects the SLD coefficient. Absorb any original reference into the channels for this argument.

First, allowing all Hermitian H in (3) is equivalent to the traceless formulation: write $H=H_0+c\rho$, where $\operatorname{Tr} H_0=0$ and $c=\operatorname{Tr} H$. Since $\Omega_A^{-1}(A)=I$ and both channels preserve trace, the added square and cross terms agree after either channel.

3.1 From SLD to relative entropy

The SLD inverse divides an off-diagonal entry by the sum of its two baseline eigenvalues. Put the two states in different reference blocks: this makes the sum involve one eigenvalue of each state. Varying their relative block weights supplies the whole resolvent family.

For $s>0$ let $f_s(x)=(x-1)^2/(x+s)$. For positive definite A,B define

$$D_{f_s}(A\|B)=\langle A-B,(L_A+sR_B)^{-1}(A-B)\rangle_{\mathrm{HS}}, \tag{6}$$

where $L_A(X)=AX$ and $R_B(X)=XB$. In mixed eigenbases this is

$$\sum_{i,j}\frac{(a_i-b_j)^2}{a_i+sb_j}|\langle u_i,v_j\rangle|^2.$$

For faithful states ρ,σ, set $X=\rho-\sigma$ and append a reference qubit:

$$\Theta_s=\frac{1}{1+s}\begin{pmatrix}\rho & 0\\ 0 & s\sigma\end{pmatrix},\qquad H_X=\begin{pmatrix}0 & X\\ X & 0\end{pmatrix}.$$

Solving the off-diagonal Sylvester equations yields

$$g_{\Theta_s}(H_X)=4(1+s)D_{f_s}(\rho\|\sigma). \tag{7}$$

The same identity holds after either channel. Complete SLD comparison therefore compares every D_{f_s}.

For clarity, the precise positive integral needed here is

$$x\log x-(x-1)=\int_0^\infty \frac{(x-1)^2}{x+s}\,\frac{s}{(1+s)^2}\,ds. \tag{8}$$

It follows by elementary integration, or by differentiating twice in x and matching the value and first derivative at $x = 1$. For states the affine term cancels, so applying the scalar spectral formula and Tonelli's theorem gives

$$D(A\|B) = \int_0^\infty D_{f_s}(A\|B)\frac{s}{(1+s)^2}\,ds.$$

This includes the usual infinite support values. Indeed the scalar perspective $h_s(a,b) = (a-b)^2/(a+sb)$, with $h_s(0,0) = 0$, is continuous on the nonnegative quadrant. Functional calculus for the commuting positive operators L_A, R_B shows that each D_{f_s} is continuous on positive semidefinite pairs in fixed dimension. Common faithful regularization extends its comparison to boundary pairs, and the nonnegative integral then gives the extended-valued entropy comparison. This proves $\succeq^{\mathrm{c}}_{\mathrm{SLD}} \Rightarrow \succeq^{\mathrm{c}}_{D}$.

3.2 A saturated reference pair

An arbitrary entropy expansion contains an uncontrolled zeroth-order channel difference. To remove it, let numerator and denominator have the same system factor at the expansion point. Their relative entropy then lives entirely in the reference and is independent of the channel. We perturb only the denominator, through coherences between one centre coordinate and several leaf coordinates.

Suppose $\mathcal{P} \succeq^{\mathrm{c}}_{D} \mathcal{Q}$. Fix a faithful ρ and Hermitian H on the input, including an arbitrary original reference. On the new reference choose a centre e_0 and N leaves, and set

$$W = \mathrm{diag}(1, \lambda_1, \ldots, \lambda_N), \quad \lambda_k > 0, \qquad \tau = vv^*, \quad L = e_0 w^* + w e_0^*, \tag{9}$$

where v, w are leaf-supported and $\|v\| = 1$. Let $Z = \mathrm{Tr}\, W$ and consider

$$\alpha = \rho \otimes \tau, \qquad \beta(\varepsilon) = \frac{\rho \otimes W + \varepsilon H \otimes L}{Z}. \tag{10}$$

The denominator has trace one because $\mathrm{Tr}\, L = 0$, and it is positive definite for sufficiently small $|\varepsilon|$ of either sign. The two auxiliary marginals at zero are deliberately different: $\alpha_{\mathrm{aux}} = \tau$, whereas $\beta(0)_{\mathrm{aux}} = W/Z$.

For either channel let $A = \Phi(\rho)$ and $K = \Phi(H)$ on its effective support. At $\varepsilon = 0$,

$$D(A \otimes \tau \| A \otimes W/Z) = D(\tau \| W/Z).$$

Thus the source-minus-target entropy difference is nonnegative near zero and vanishes at zero. Its second derivative is nonnegative. Conjugating the auxiliary reference by $\mathrm{diag}(-1, 1, \ldots, 1)$ also shows that each entropy along the path is even.

3.3 The denominator Hessian and realizable filters

The leaf weights determine a finite rational function F. Its squared modulus weights the Hessian kernels $(a_j + a_i t)^{-1}$. If that weight concentrates at $t = 1$, the Hessian becomes precisely the SLD form. The calculation below makes this spectral selection explicit.

For $B > 0$ and Hermitian Y, the logarithm resolvent formula gives

$$-\frac{d^2}{d\varepsilon^2}\log(B + \varepsilon Y)\bigg|_0 = 2\int_0^\infty (B+u)^{-1} Y (B+u)^{-1} Y (B+u)^{-1}\,du. \tag{11}$$

Define the finite resolvent sum

$$F(t) = w^*(W_{\mathrm{leaf}} + tI)^{-1} v.$$

In an eigenbasis of A, the two exterior resolvents in (11) act on leaves, and the middle resolvent acts on the centre. For the ordered system indices (i, j) the contribution is

$$2|K_{ij}|^2 a_i \int_0^\infty \frac{|F(u/a_i)/a_i|^2}{a_j + u}\, du.$$

Substituting $u = a_i t$ proves the exact identity

$$\frac{d^2}{d\varepsilon^2} D\left(A \otimes \tau \middle\| \frac{A \otimes W + \varepsilon K \otimes L}{Z}\right)\Big|_0 = 2\sum_{i,j} |K_{ij}|^2 \int_0^\infty \frac{|F(t)|^2}{a_j + a_i t}\, dt. \tag{12}$$

The normalization contributes only $\log Z$ and does not affect the derivative. The integrals converge because F is bounded at zero and $O(t^{-1})$ at infinity.

Every finite simple-resolvent sum $F(t) = \sum_k c_k/(t + \lambda_k)$ is realizable: take $v_k = N^{-1/2}$ and $w_k = \sqrt{N}\,\overline{c_k}$. Large but finite coefficients merely make the admissible interval of ε smaller.

Lemma 3.2 (Finite filters concentrating at one)**.** *For $m \geq 1$ define $\lambda_j = 1 + j/m^3$ for $0 \leq j \leq 2m$ and*

$$F_m(t) = \frac{t^m}{\prod_{j=0}^{2m}(t + \lambda_j)}, \qquad I_m = \int_0^\infty F_m(t)^2\, dt.$$

These functions are realizable finite resolvent sums with rational poles and coefficients, and for $a, b > 0$,

$$\frac{1}{I_m} \int_0^\infty \frac{F_m(t)^2}{b + at}\, dt \longrightarrow \frac{1}{a + b}. \tag{13}$$

Proof. Partial fractions give

$$F_m(t) = \sum_{j=0}^{2m} \frac{c_j}{t + \lambda_j}, \qquad c_j = \frac{(-\lambda_j)^m}{\prod_{k \neq j}(\lambda_k - \lambda_j)} \in \mathbb{Q}.$$

Put $f_m(t) = t^m/(1 + t)^{2m+1}$. Since $\log(1 + x) \leq x$,

$$\exp(-2/m - 1/m^2) \leq F_m(t)/f_m(t) \leq 1 \quad (t > 0).$$

The concentration mechanism is visible already in f_m: the factor $[t/(1 + t)^2]^m$ has its unique maximum at $t = 1$. The nearby distinct poles make the filter a finite sum of realizable resolvents without changing that concentration. The normalized density proportional to f_m^2 is beta-prime with parameters $(2m + 1, 2m + 1)$. Direct beta-integral evaluation gives its mean $(2m + 1)/(2m)$ and variance

$$\frac{(2m + 1)(4m + 1)}{(2m - 1)(2m)^2}.$$

It therefore converges in probability to one. The displayed multiplicative bounds show that the normalized F_m^2 densities differ from these densities by a total-variation error tending to zero. Applying this to the bounded continuous function $(b + at)^{-1}$ proves (13). □

Apply the entropy hypothesis to (10) with each F_m. The difference of (12) is nonnegative. Divide by I_m and take the limit, using the finitely many output eigenvalues. The result is exactly $g_{\mathcal{P}(\rho)}(\mathcal{P}(H)) \geq g_{\mathcal{Q}(\rho)}(\mathcal{Q}(H))$. The original reference was arbitrary, proving $\succeq_D^{\mathrm{c}} \Rightarrow \succeq_{\mathrm{SLD}}^{\mathrm{c}}$. Every auxiliary reference used has finite dimension $2m + 2$. If one assumes entropy comparison only for faithful numerators, replace τ by $(1 - \zeta)\tau + \zeta I/(N + 1)$ and pass to $\zeta \downarrow 0$; saturation persists and the Hessian is linear and continuous in the numerator.

4 Why the environment reverses the order

A Stinespring isometry loses no information; loss occurs only when one output is discarded. Purification geometry measures that loss by the freedom to change the discarded system without changing the receiver's state. For the SLD metric, the resulting least-squares identity turns an input score perturbation into a phase perturbation at the environment, as in the metrological work of Faist et al. [10].

There is a subtle comparison issue. To subtract the identities for two channels, their dual experiments must be identical. Choosing a different purifier after each channel would not establish an order. We construct the dual state and tangent before either channel is applied. The original reference becomes part of the system being purified, while the purifier becomes the new reference. This exchange is why complete comparison is the natural hypothesis.

4.1 A common dual experiment

For $\rho > 0$ define

$$\Omega_\rho(H) = \frac{\rho H + H\rho}{2}, \qquad g_\rho(X) = \operatorname{Tr}[X\Omega_\rho^{-1}(X)].$$

At a possibly singular positive matrix A, use the supported extension

$$g_A(B) = \sum_{i,j:a_i+a_j>0} \frac{2|B_{ij}|^2}{a_i + a_j}, \tag{14}$$

provided the kernel-to-kernel block of B is zero; otherwise set $g_A(B) = +\infty$. Support-to-kernel entries are retained.

Lemma 4.1 (Purification identity)**.** *Let $\Phi : A \to B$ have genuine complement $\Phi^c : A \to E$. Let $\rho_{RA} > 0$, $X_{RA} = X_{RA}^*$ and $\operatorname{Tr} X = 0$. Set $H = \Omega_\rho^{-1}(X)$ and choose a purification $|\psi\rangle_{RAF}$ of ρ, where F is a copy of RA. Define*

$$|\psi(u)\rangle = (e^{iuH/2} \otimes I_F)|\psi\rangle,$$
$$\Gamma = \operatorname{Tr}_R |\psi\rangle\langle\psi|, \qquad \xi = \frac{i}{2}\operatorname{Tr}_R[H \otimes I_F, |\psi\rangle\langle\psi|].$$

The pair (Γ, ξ) is the same for every channel Φ on A, and

$$g_{\Phi_R(\rho)}(\Phi_R(X)) = g_\rho(X) - g_{\Phi_F^c(\Gamma)}(\Phi_F^c(\xi)). \tag{15}$$

Here $\Phi_R = \Phi \otimes \mathrm{id}_R$ and $\Phi_F^c = \Phi^c \otimes \mathrm{id}_F$.

Proof. Let $V : A \to BE$ be the isometry, and put

$$|\Psi\rangle = (I_R \otimes V \otimes I_F)|\psi\rangle, \qquad |\dot\Psi\rangle = (I_R \otimes V \otimes I_F)(H \otimes I_F)|\psi\rangle/2.$$

The real score flow has output tangent $\Phi_R(X)$, and

$$4\|\dot\Psi\|^2 = \operatorname{Tr}(\rho H^2) = g_\rho(X), \qquad \langle\Psi|\dot\Psi\rangle = \operatorname{Tr}(\rho H)/2 = 0.$$

For the partition $RB : EF$, set

$$A_0 = \operatorname{Tr}_{RB} |\Psi\rangle\langle\Psi| = \Phi_F^c(\Gamma), \qquad B_0 = i\operatorname{Tr}_{RB}(|\dot\Psi\rangle\langle\Psi| - |\Psi\rangle\langle\dot\Psi|) = \Phi_F^c(\xi).$$

The horizontal-purification variational formula for the SLD metric gives

$$g_{\Phi_R(\rho)}(\Phi_R(X)) = \min_{J=J^*} 4\|\dot\Psi - i(I_{RB} \otimes J)\Psi\|^2. \tag{16}$$

Here is a direct justification of this standard factor-four relation. Write a purification as a coefficient matrix P with $PP^* = \sigma$, where $\sigma = \Phi_R(\rho)$ is faithful on its effective support. A tangent matrix Q satisfies $QP^* + PQ^* = Y$, the prescribed output derivative. With $H_\sigma = \Omega_\sigma^{-1}(Y)$, the particular tangent $Q_0 = H_\sigma P/2$ satisfies that constraint. For every difference K with $KP^* + PK^* = 0$,

$$\operatorname{Re}\operatorname{Tr}(Q_0^* K) = \tfrac{1}{4}\operatorname{Tr}\big[H_\sigma(KP^* + PK^*)\big] = 0.$$

Thus Q_0 minimizes the squared norm and $4\|Q_0\|^2 = g_\sigma(Y)$. In a Schmidt basis $P = [\sqrt{\sigma}\ \ 0]$, every such K has the form iPJ with J Hermitian: the first block follows from $\sigma^{-1/2}K_1$ being anti-Hermitian, and the second block fixes an arbitrary off-diagonal block of J. This proves (16) with the actual purifier, including redundant dimensions. Expanding its right side gives

$$g_\rho(X) + 4\operatorname{Tr}(A_0 J^2) + 4\operatorname{Tr}(B_0 J).$$

The normal equation is $\Omega_{A_0}(J) = -B_0/2$. Completing the square on support pairs therefore gives $g_\rho(X) - g_{A_0}(B_0)$, as asserted. Components of J wholly inside $\ker A_0$ do not affect the value.

The original output $\Phi_R(\rho)$ is faithful on the fixed effective support of $\Phi_R(I)$, so its real score flow has constant support. Although Γ may be singular, $\Gamma(u) = \operatorname{Tr}_R |\psi(u)\rangle\langle\psi(u)|$ has constant rank: its nonzero spectrum is the spectrum of the faithful original-R marginal. Similarly, $\Phi_F^c(\Gamma(u))$ has the nonzero spectrum of $\Phi_R(e^{iuH/2}\rho e^{-iuH/2})$, which has fixed rank. Thus the phase tangent has zero kernel-to-kernel block, and the supported quadratic form in (14) is finite. □

Theorem 4.2 (Complement reversal)**.** *For genuine complements,*

$$\Phi \succeq^{\mathrm{c}}_{\mathrm{SLD}} \Psi \quad \Longleftrightarrow \quad \Psi^c \succeq^{\mathrm{c}}_{\mathrm{SLD}} \Phi^c. \tag{17}$$

By Theorem 3.1, the same statement holds for each equivalent order, in particular with $\succeq^{\mathrm{c}}_D$ in place of $\succeq^{\mathrm{c}}_{\mathrm{SLD}}$.

Proof. To extend a faithful-input SLD inequality to a dual pair from Lemma 4.1, take

$$\Gamma_\varepsilon = (1-\varepsilon)\Gamma + \varepsilon I_{AF}/\dim(AF), \qquad \xi_\varepsilon = (1-\varepsilon)\xi.$$

Compress each output to its fixed effective support. Spectral perturbation in (14), equivalently the inverse-Jordan quadratic variational formula, yields

$$g_{\Lambda(\Gamma_\varepsilon)}(\Lambda(\xi_\varepsilon)) \longrightarrow g_{\Lambda(\Gamma)}(\Lambda(\xi)).$$

The absent kernel-to-kernel tangent block prevents a divergent term; support-to-kernel terms retain denominators tending to a positive eigenvalue. Hence complete-order inequalities pass to this supported limit.

Assume $\Phi \succeq^{\mathrm{c}}_{\mathrm{SLD}} \Psi$. Apply (15) to the channels Φ^c, Ψ^c on an arbitrary faithful primal pair (ρ, X). Their double complements are Φ, Ψ up to output isometries. The common dual pair and the extended assumption give

$$g_{\Phi_R^c(\rho)}(\Phi_R^c(X)) = g_\rho(X) - g_{\Phi_F(\Gamma)}(\Phi_F(\xi)) \le g_\rho(X) - g_{\Psi_F(\Gamma)}(\Psi_F(\xi)) = g_{\Psi_R^c(\rho)}(\Psi_R^c(X)).$$

This proves the forward implication on all primal pairs; surjectivity of the dual construction is unnecessary. Applying the same implication again to Ψ^c, Φ^c proves the reverse. Output isometries preserve the quadratic forms. Finally use Theorem 3.1 on both sides to obtain the Umegaki version. □

The original reference R and the dual purifier F are different systems. No state-dependent polar unitary is identified between different baselines. The dual identity concerns the SLD quadratic form, and is not an identity for the Petz order-two or right-logarithmic-derivative endpoint.

5 From a nonlocal game to a gapped polynomial

The local description of the order is now useful in a different way: it tells us which operator expressions a measurement can test. After vectorization, a Sylvester operator is a sum of left and right actions. This resembles the two players of a nonlocal game, but each accessible letter mixes both actions with the same, generally unequal weights. We must recover the players from these mixtures, rather than insert arbitrary game observables by assumption.

We first describe the accessible domain. The Pauli anchor identifies its balanced Bell sector; smoothing then recovers local measurements near that sector. The result is one polynomial whose sign detects the game value and whose coefficients are computable from the game table.

Notation and dependencies. The following guide collects the operators introduced in the reduction. Here R, S denote positive baseline matrices; in particular, R no longer labels a reference system. All letters in an instance share these same baselines.

Symbols	Role in the construction
$R, S;\ B_A, D, Z_A$	R, S are faithful, possibly unnormalized physical–reference baselines. B_A is the vectorized Sylvester denominator for effect A; $D = B_I$ and $Z_A = D^{-1/2} B_A D^{-1/2}$ normalize it (18).
p, q, j	The two weight operators and their imbalance: $p^2 + q^2 = I$, $j = p^2 - q^2$. The anchor controls their imbalance on the tested vector (22).
$x_i, y_i;\ a_g, b_g$	Oppositely oriented local Pauli representations and local payload effects, with $i \in \{X, Y, Z\}$ and $g = (x, a)$ a question–answer label (23).
$T_i, T_{i,g};\ P_g, Q_g$	Accessible weighted mixtures of the Pauli and payload operators; P_g, Q_g extract the two players' payloads and equal a_g, b_g in the honest balanced model (21), (31).
$\Delta;\ E_{\mathrm{op}}, H_{\mathrm{op}};\ F$	Anchor defect; anchor and question-normalization penalties; penalized game polynomial (43)–(46). Here F is unrelated to the reference filter of Section 3.
$p(Z), K(R, S)$	The same game polynomial in the letters Z_A, then its inverse Sylvester kernel (47). The polynomial $p(Z)$ is distinct from the weight operator p.
$\Gamma_{\bar{R}}, \mathcal{C};\ (G_h)$	Fisher difference on the selected cross-block tangent space; additional block-diagonal measurements extend its sign to all inputs and tangents (Sections 6.2–6.3).

Lemma 5.1 and local smoothing lead to the gapped polynomial of Proposition 5.3. Lemma 6.1 converts its gap into a kernel gap, and Proposition 6.2 converts that gap into complete SLD dominance; a strict finite negative witness survives both conversions. Theorem 3.1 then transfers the undecidability to the equivalent entropy orders. Complement reversal (Theorem 4.2) is independent of this reduction. Fixed-error inapproximability additionally needs the dimension-independent witness margin of Section 8.2, Lemma 8.4, and the additivity in Proposition 8.2.

The external game theorem. From a Turing machine and its input one can effectively construct a finite rational two-player game G such that

$$\begin{aligned} &\text{halting:} && G \text{ has a finite-dimensional perfect PCC strategy,} \\ &\text{nonhalting:} && \omega^*(G) \le \tfrac{1}{2}. \end{aligned}$$

Here ω^* is the supremum over finite-dimensional tensor-product strategies. This is Theorem 12.6 of Ji et al. [1], arXiv v3, printed p. 172. A PCC strategy is projective and consistent: every answer projector M in either player's family satisfies $(M \otimes I)\psi = (I \otimes M)\psi$; its additional question-wise commutation property is not needed here. The finite verifier can be expanded

to a rational game table. Efficiency of this expansion or of our subsequent reduction is not asserted. For exact-order hardness only a strategy scoring above 7/8 is needed. Perfect consistency is used later to obtain a margin independent of the unknown strategy dimension.

5.1 The accessible tensor–Jordan domain

Fix physical dimension h. For a Hermitian physical effect A and $R > 0$ on $\mathbb{C}^h \otimes \mathbb{C}^n$, write

$$\mathcal{A}_A(R) = \mathrm{Tr}_h[(A \otimes I_n)R].$$

For $S > 0$ on $\mathbb{C}^h \otimes \mathbb{C}^m$ define, in a fixed tensor-vectorization convention,

$$B_A(R,S) = \mathcal{A}_A(R) \otimes I_m + I_n \otimes (\mathcal{A}_A(S))^{\mathsf{T}}, \quad D = B_I(R,S), \quad Z_A = D^{-1/2} B_A D^{-1/2}. \tag{18}$$

The baselines R, S need not have trace one; we normalize the final metric witness using homogeneity. The transpose is on the second reference output. A fixed tensor-factor swap turns this convention into column vectorization of (2). If $\alpha I \le A \le (1-\alpha)I$, then

$$\alpha I \le Z_A \le (1-\alpha)I, \qquad B_{I-A} = D - B_A. \tag{19}$$

All effects in a tuple use the *same* R, S, D. The shared two-factor form is a constraint, not a notational convenience. A proof on arbitrary unrelated operator tuples would not settle the comparison needed here.

5.1.1 The physical Pauli and payload algebra

Pad the game to M questions and K outcomes on each side. Zero-probability questions and duplicated outcomes preserve its value. Put $N = MK$, $h = 2N$, and let e_g, $g = (x,a)$, be the minimal projections of $\mathbb{C}^N$. Inside M_h use $M_2 \otimes \mathbb{C}^N$. With the usual Pauli matrices $\sigma_X, \sigma_Y, \sigma_Z$ and $\tau = 1/2$, take

$$A_i = \frac{I + \tau\sigma_i \otimes I}{2}, \qquad A_{i,g} = \frac{I + \tau\sigma_i \otimes e_g}{2}. \tag{20}$$

These rational complex effects and their complements lie between $I/4$ and $3I/4$. Define centered contractions

$$T_i = (2Z_{A_i} - I)/\tau, \qquad T_{i,g} = (2Z_{A_{i,g}} - I)/\tau. \tag{21}$$

They are selfadjoint contractions by positivity of the normalized maps.

Put $W = \mathrm{Tr}_h R$, $V = (\mathrm{Tr}_h S)^{\mathsf{T}}$, and on the two reference factors set

$$r = W \otimes I, \quad s = I \otimes V, \quad p = r^{1/2}(r+s)^{-1/2}, \quad q = s^{1/2}(r+s)^{-1/2}, \quad j = (r-s)(r+s)^{-1}. \tag{22}$$

The Choi correspondence provides UCP maps of the physical algebra. Explicitly, the left map is $B \mapsto W^{-1/2}\,\mathrm{Tr}_h[(B^{\mathsf{T}} \otimes I)R]W^{-1/2}$; the right map is $B \mapsto V^{-1/2}(\mathrm{Tr}_h[(B \otimes I)S])^{\mathsf{T}}V^{-1/2}$. Both are completely positive and unital. The left Pauli orientation is therefore transposed and the right orientation is not.

After common local Stinespring dilations of the *entire* algebra, write the exact opposite Pauli representations as x_i, y_i, and the local payload projections as a_g, b_g. They commute with their respective Pauli representations and form global POVMs. The normalized variables have the form

$$T_i = p x_i p + q y_i q, \qquad T_{i,g} = p x_i a_g p + q y_i b_g q. \tag{23}$$

For general UCP compressions this formula is used through the positive-weight dilation limit proved below. It is not a claim that arbitrary decomposable compressions satisfy the same tensor-local relations.

5.1.2 Honest finite strategy realization

Given local question-wise POVMs $M_{a|x}$ and $N_{b|y}$, set $a_{x,a} = M_{a|x}/M$, $b_{y,b} = N_{b|y}/M$. Define UCP maps on $\bigoplus_{g=1}^{N} M_2$ by

$$\Phi_A((Z_g)_g) = \sum_g Z_g \otimes a_g, \qquad \Phi_B((Z_g)_g) = \sum_g Z_g \otimes b_g.$$

Their Choi baselines have identity reference marginals. Thus $p = q = I/\sqrt{2}$ and

$$T_i = (x_i + y_i)/2, \qquad T_{i,g} = (x_i a_g + y_i b_g)/2, \quad x_i = \sigma_i^{\mathsf{T}}, \quad y_i = \sigma_i, \tag{24}$$

with the strategy-space tensor factors understood. The baselines may be singular, but $D = 2I$ remains invertible. These honest models belong to the positive semidefinite closure of the domain. Strict negativity will later be preserved by an explicit faithful perturbation.

5.2 The Pauli anchor: recovering two local systems

The three Pauli directions distinguish a genuine two-player mixture from a one-sided representation. In the balanced model the opposite Pauli actions agree on a Bell vector. Their averages square to the identity on that vector, yet their commutators annihilate it. Both tests are needed. With scalar weights $0 \le t \le 1$, take $|\Phi_2\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Then

$$T_i = tx_i + (1-t)y_i, \qquad [T_i, T_j]|\Phi_2\rangle = 2\mathrm{i}(1-2t)y_k|\Phi_2\rangle \quad ((i,j,k) \text{ cyclic}).$$

The involution tests alone pass on $|\Phi_2\rangle$ for every t; the commutator tests force $t = 1/2$. The anchor below makes the same conclusion robust for operator-valued weights that need not commute with their local Pauli actions. Crucially, its constants do not depend on a smallest marginal eigenvalue.

For exact opposite Pauli representations in (23) and any vector v, set

$$\begin{aligned} e_i(v) &= \langle v, (I - T_i^2)v\rangle, & e(v) &= \sum_i e_i(v), \\ c(v) &= \sum_{(i,j)\ \text{cyclic}} \|[T_i, T_j]v\|^2, & \delta(v) &= e(v) + c(v). \end{aligned} \tag{25}$$

Let $P = |\Phi_2\rangle\langle\Phi_2| \otimes I$ be the Bell projector for the two opposite Pauli representations, and $Q = I - P$.

Lemma 5.1 (Anchor rigidity)**.** *For every vector v,*

$$\|Qv\|^2 \le 10\delta(v), \qquad \|jv\|^2 \le 10\delta(v), \qquad \|(p - 2^{-1/2})v\|^2, \|(q - 2^{-1/2})v\|^2 \le 5\delta(v). \tag{26}$$

Also

$$\|T_i v - x_i v\| \le 7\sqrt{\delta(v)}, \qquad \|jT_i v\| \le 2\sqrt{\delta(v)}. \tag{27}$$

The same estimates hold for arbitrary UCP maps after fixed common local Stinespring isometries, independently of dimensions and marginal condition numbers.

The proof uses an inverse-root change of variables to compare two weighted Pauli vectors. Its multiplier is a contraction even when a marginal is nearly singular. The Pauli algebra then forces both vectors into the same Bell sector and forces their weights to agree. This avoids the condition-number loss that a direct inversion would introduce. Appendix A.1 gives the full estimates, including the passage from representations to UCP maps.

5.2.1 Transport of words and a single payload

Define honest balanced anchors $K_i = (x_i + y_i)/2$. On a Bell vector, $K_i P v = x_i P v$ and $K_i x_j P v = \delta_{ij} P v$. The commutator and anticommutator residual bounds imply, for $i \neq j$,

$$\|T_i T_j v\| \leq \frac{\|[T_i, T_j]v\| + \sqrt{e_i(v)} + \sqrt{e_j(v)}}{2} \leq \sqrt{\delta(v)}.$$

For $i = j$, $\|(T_i^2 - I)v\| \leq \sqrt{e_i(v)}$. Reduce pairs from the right in a word w of length d. Each replacement costs at most $\sqrt{\delta(v)}$, and all unreduced prefixes are contractions. The corresponding K word makes exactly the same reductions on Pv. Consequently

$$\|T_w v - K_w P v\| \leq \big(\lfloor d/2 \rfloor + \sqrt{10} + 7\mathbf{1}_{d \text{ odd}}\big)\sqrt{\delta(v)}. \tag{28}$$

The same reductions and (27) give

$$\|j T_w v\| \leq b_d \sqrt{\delta(v)}, \qquad b_d = \lfloor d/2 \rfloor + \sqrt{10} + 2. \tag{29}$$

Fix a local payload effect a_g, b_g, and put $K_{H,g} = (x_H a_g + y_H b_g)/2$. Moving the outside p, q factors to the bra in $\langle T_u w, T_{H,g} T_t v\rangle$, replacing them by $1/\sqrt{2}$ using (29), and then using (28), proves, for $|u|, |t| \leq d$,

$$\begin{aligned} &|\langle T_u w, T_{H,g} T_t v\rangle - \langle K_u P w, K_{H,g} K_t P v\rangle| \\ &\quad \leq C_d\big(\|w\|\sqrt{\delta(v)} + \|v\|\sqrt{\delta(w)}\big), \qquad C_d = \sqrt{2} b_d + \lfloor d/2 \rfloor + \sqrt{10} + 7. \end{aligned} \tag{30}$$

This proof uses only local contractions and tested word actions.

Choose cyclic indices $(K, L, H) = (X, Y, Z)$ and define Hermitian extractors

$$\begin{aligned} L_H &= T_H + \mathrm{i}[T_K, T_L], & R_H &= T_H - \mathrm{i}[T_K, T_L], \\ U_g &= T_{H,g} + \mathrm{i}[T_K, T_{L,g}], & V_g &= T_{H,g} - \mathrm{i}[T_K, T_{L,g}], \\ P_g &= (L_H U_g + U_g L_H)/2, & Q_g &= (R_H V_g + V_g R_H)/2. \end{aligned} \tag{31}$$

They have degree at most four, operator norm at most nine, and monomial coefficient absolute sum at most nine. Every monomial has exactly one raw payload occurrence. In the honest balanced model, opposite chirality gives $L_H = x_H$, $R_H = y_H$, $U_g = x_H a_g$, $V_g = y_H b_g$, so $P_g = a_g$, $Q_g = b_g$ as operators.

Since a_g, b_g commute with the Bell projector, (30) removes the Bell projections at an extra cost at most $\sqrt{10}\|w\|\sqrt{\delta(v)}$. Using $d = 3$ and $9C_3 + \sqrt{10} < 200$ gives

$$|\langle w, (P_g - a_g)v\rangle| \leq 200\big(\|w\|\sqrt{\delta(v)} + \|v\|\sqrt{\delta(w)}\big), \tag{32}$$

and the same estimate for $Q_g - b_g$. This is initially a weak matrix-element estimate, not yet an action-norm estimate.

5.3 From approximate payloads to a valid strategy

The anchor has located the two players, but has not yet produced a strategy. The extracted P_g, Q_g are polynomials, not positive measurement effects, and (32) controls a matrix element only when both test vectors have low anchor energy. Taking a supremum over arbitrary test vectors would therefore be invalid.

We smooth the test vector first. The smoothing must also preserve locality: projecting onto a global low-energy subspace could entangle the players' operations. A positive contraction that is an average of product unitaries solves both problems. Its powers suppress high-energy test vectors, while averaging conjugates of the local payloads gives genuine local POVMs. Only the root and its finitely many payload descendants need to have small anchor energy.

5.3.1 A smoothing contraction from local unitaries

The balance operator measures the mismatch between the two marginal weights. A Gaussian function of their logarithmic ratio suppresses that mismatch. Its Fourier representation is useful for a different reason: each Fourier component acts locally on the two players. Averaging these local changes therefore smooths the extracted operators while preserving the tensor-product form of a strategy.

For strictly positive exact-representation weights define

$$\Lambda = \log r - \log s, \quad G = e^{-\Lambda^2/2}, \quad x_0 = I, \qquad M_0 = \frac{P + \sum_{i=0,X,Y,Z} x_i G x_i}{5}. \tag{33}$$

It is a positive contraction and an average of product-local unitaries. To see the latter, $P = (I + \sum_i x_i y_i)/4$ is such an average, and the Gaussian Fourier identity expresses G as the average of $r^{\mathrm{i}t} s^{-\mathrm{i}t}$ with a standard normal parameter t. Conjugating by x_i preserves the property. Hence for every integer $n \geq 1$,

$$M_0^n = \mathbb{E}_\omega [U_\omega \otimes V_\omega] \tag{34}$$

for local unitaries on the two finite spaces. This average is an analytic existence device; it is not part of the symbolic output.

Set

$$E_M(v) = \langle v, (I - M_0)v \rangle, \qquad E_0(v) = \|Qv\|^2 + \sum_{i=0,X,Y,Z} \|j x_i v\|^2.$$

Since $j = \tanh(\Lambda/2)$, the scalar bounds

$$\tanh^2(t/2) \leq 1 - e^{-t^2/2} \leq 8 \tanh^2(t/2)$$

imply

$$E_0(v)/5 \leq E_M(v) \leq 8E_0(v)/5. \tag{35}$$

For the lower scalar bound use $\cosh(t/2) \leq e^{t^2/8}$; for the upper use $\tanh(|t|/2) \geq |t|/4$ when $|t| \leq 1$ and $\tanh(1/2)^2 > 1/5$ otherwise.

Lemma 5.2 (Energy comparison)**.** *Uniformly over all the exact-representation instances,*

$$E_M(v) \leq 421\delta(v), \qquad \delta(v) \leq 3000 E_M(v). \tag{36}$$

Therefore

$$\delta(M_0^n w) \leq \frac{3000}{2n+1} \|w\|^2, \qquad \|v - M_0^n v\| \leq \sqrt{n E_M(v)}. \tag{37}$$

The equivalence of energies is proved in Appendix A.2. The decay in (37) is ordinary spectral filtering: for $0 \leq t \leq 1$, $t^{2n}(1-t) \leq (2n+1)^{-1}$ and $(1-t^n)^2 \leq n(1-t)$. Thus smoothing makes arbitrary test vectors good and changes an already good vector only slightly.

5.3.2 From weak transport to local action transport

Use the same averaging (34) for every payload on its party:

$$a_g^{(n)} = \mathbb{E}_\omega U_\omega a_g U_\omega^*, \qquad b_g^{(n)} = \mathbb{E}_\omega V_\omega b_g V_\omega^*. \tag{38}$$

These are local positive contractions and each family sums to the identity. Applying (32) with w replaced by $M_0^n w$ and using (37) gives by duality

$$\|M_0^n (P_g - a_g) v\| \leq 200\sqrt{\delta(v)} + 200\sqrt{3000/(2n+1)}\|v\|.$$

Also the exact average identity

$$\mathbb{E}_\omega \|(U_\omega \otimes V_\omega)v - v\|^2 = 2\langle v, (I - M_0^n)v\rangle \le 2nE_M(v)$$

shows $\|M_0^n a_g v - a_g^{(n)} v\| \le \sqrt{2nE_M(v)}$. Combining with (37) on $P_g v$ proves

$$\begin{aligned}\|P_g v - a_g^{(n)} v\| \le \sqrt{421n\delta(P_g v)} + \sqrt{842n\delta(v)} \\ + 200\sqrt{\delta(v)} + 200\sqrt{3000/(2n+1)}\|v\|.\end{aligned} \tag{39}$$

The same estimate holds for $Q_g, b_g^{(n)}$. For a unit root z let

$$E(z) = \delta(z) + \sum_g \delta(P_g z) + \sum_g \delta(Q_g z). \tag{40}$$

For every $n \ge 1$, each action error in (39) is at most

$$\kappa_n = C_s(\sqrt{n}\sqrt{E(z)} + n^{-1/2}), \qquad C_s = 12000. \tag{41}$$

This follows directly by bounding the four coefficients in (39); it requires no small-energy assumption.

5.3.3 Question normalization without an invertibility assumption

Although $a_g^{(n)}$ form a global POVM, the scaled effects $Ma_{x,a}^{(n)}$ need not sum to I for each question. Put

$$C_x = \sum_a M a_{x,a}^{(n)}, \qquad D_y = \sum_b M b_{y,b}^{(n)}.$$

They satisfy $0 \le C_x, D_y \le MI$. Their square-root measurement columns have norms at most $\sqrt{M}$. For one such column C with $C^*C = Q$, extend its polar partial isometry to an isometry N by filling $\ker Q$ in one chosen outcome. This is possible within the same column space because all $\sqrt{E_a}$ annihilate $\ker Q$. Then

$$C = N\sqrt{Q}, \qquad \|(C - N)z\| = \|(\sqrt{Q} - I)z\| \le \|(Q - I)z\|.$$

The POVM of N has effects $Q^{-1/2}E_aQ^{-1/2}$ on $\operatorname{supp} Q$, with the kernel added to one outcome. It is a genuine local POVM even if Q has a kernel.

Let $\tau_x = \|(C_x - I)z\|$ and $\upsilon_y = \|(D_y - I)z\|$. The difference of the joint unnormalized and normalized columns at questions x, y has norm at most $\sqrt{M}\upsilon_y + \tau_x$. The first joint column has norm at most M and the second is an isometry. For any diagonal payoff contraction, its expectation therefore changes by at most $(M+1)(\sqrt{M}\upsilon_y + \tau_x)$. Averaging with the game distribution π and applying Cauchy–Schwarz gives a score error at most

$$C_3\sqrt{H_n}, \qquad C_3 = (M+1)^2, \qquad H_n = \sum_x \pi_A(x)\tau_x^2 + \sum_y \pi_B(y)\upsilon_y^2. \tag{42}$$

Here we used $(M+1)\sqrt{M+1} \le C_3$. No scaled effect $Ma_g^{(n)}$ has been incorrectly treated as a contraction.

5.3.4 Passage back to arbitrary UCP maps

For original compressed maps, extend the weights by εI on the Stinespring complements as in Lemma 5.1. For every $\varepsilon > 0$, perform the smoothing and measurement rounding on those finite enlarged spaces. The resulting strategy is valid and its score is at most $\omega^*(G)$. Derive the scalar bound (45) below at that positive ε, eliminating its rounding errors in favor of the polynomial quantities B, E, H. For a fixed original instance, these finitely many word expectations and descendant energies converge as $\varepsilon \downarrow 0$. Passing only that scalar inequality to the limit proves it for the original compressed maps. Convergence of the rounded measurements or their normalization errors is unnecessary. The constants are independent of ε and of marginal conditioning; no compactness across dimensions or uniform rate of this auxiliary limit is used.

5.4 A game polynomial with a uniform gap

We can now assemble the polynomial. It rewards the game score and penalizes precisely the two ways an apparent strategy can be spurious: failure of the anchor on the root or a payload descendant, and failure of question-wise normalization. The rounding argument bounds the possible gain from either defect by its square root; sufficiently large quadratic penalties absorb that gain.

Let G have rational question distribution π and rational payoff $0 \le V(a,b|x,y) \le 1$. Define the positive polynomial

$$\Delta = \sum_i (I - T_i^2) + \sum_{(i,j)\ \text{cyclic}} [T_i, T_j]^*[T_i, T_j].$$

Its quadratic form is δ. Define

$$E_{\mathrm{op}} = \Delta + \sum_g P_g \Delta P_g + \sum_g Q_g \Delta Q_g,$$

$$H_{\mathrm{op}} = \sum_x \left(M \sum_a P_{x,a} - I \right)^2 + \sum_y \left(M \sum_b Q_{y,b} - I \right)^2, \tag{43}$$

and put $E(z) = \langle z, E_{\mathrm{op}} z\rangle$, $H(z) = \langle z, H_{\mathrm{op}} z\rangle$. The first has degree at most twelve, the second at most eight. Both are positive on the actual domain. The Hermitian game-score polynomial is

$$B = M^2 \sum_{x,y,a,b} \pi(x,y) V(a,b|x,y) \frac{P_{x,a}Q_{y,b} + Q_{y,b}P_{x,a}}{2}. \tag{44}$$

In the honest model (24), on a Bell anchor tensored with any strategy vector, every extracted descendant keeps its Bell factor. Thus $E = H = 0$, and $\langle z, Bz\rangle$ is exactly the strategy's game score.

For an arbitrary actual instance and unit z, the two action approximations with error κ_n and $\|P_g\|, \|Q_h\| \le 9$ show

$$|\langle P_g z, Q_h z\rangle - \langle a_g^{(n)} z, b_h^{(n)} z\rangle| \le 10\kappa_n.$$

There are K^2 outcome pairs for each question pair. The discrepancy between (44) and the unnormalized local score is at most $10M^2K^2\kappa_n$. Moreover Minkowski's inequality gives

$$\sqrt{H_n} \le \sqrt{H} + \sqrt{2}MK\kappa_n \le \sqrt{H} + 2MK\kappa_n.$$

Use (42) to normalize the questions. The resulting valid local strategy has score at most $\omega^*(G)$, so

$$\langle z, Bz\rangle \le \omega^*(G) + C_1 C_s \sqrt{n}\sqrt{E} + C_1 C_s/\sqrt{n} + C_3\sqrt{H}, \quad C_1 = 10M^2K^2 + 2C_3MK. \tag{45}$$

The preceding dilation argument proves this on every actual normalized domain instance.

Proposition 5.3 (Promised gapped polynomial)**.** *Given $r \in \mathbb{Q}$ and $\gamma \in \mathbb{Q}_{>0}$, one can effectively construct a rational Hermitian polynomial F of degree at most twelve such that*

1. *If $\omega^*(G) \le r - \gamma$, then $F(T) \ge (\gamma/2)I$ throughout the actual normalized tensor-Jordan domain.*
2. *If $\omega^*(G) > r + \gamma/2$, then F has a strict negative finite-dimensional witness with faithful baselines.*

Proof. Choose

$$s_0 = \max\{1, \lceil 4C_1C_s/\gamma \rceil\}, \qquad n = s_0^2, \qquad C_2 = C_1C_s s_0,$$
$$\lambda_E = \lceil 2C_2^2/\gamma \rceil, \qquad \lambda_H = \lceil 2C_3^2/\gamma \rceil,$$

and set

$$F = rI - B + \lambda_E E_{\mathrm{op}} + \lambda_H H_{\mathrm{op}}. \tag{46}$$

In the low-value case, (45) gives

$$\langle z, Fz \rangle \ge 3\gamma/4 - C_2\sqrt{E} - C_3\sqrt{H} + \lambda_E E + \lambda_H H.$$

Completing each scalar square costs at most $\gamma/8$, proving the gap.

In the high-value case choose a finite strategy scoring above $r + \gamma/2$; it exists by the definition of the supremum. Purify its state if necessary, keeping local spaces finite. Its honest Bell realization has $E = H = 0$ and $\langle z, Fz \rangle < -\gamma/2$. To make the Choi baselines faithful, let their identity marginals be fixed and replace

$$R_0 \mapsto R_\varepsilon = (1-\varepsilon)R_0 + \varepsilon(I_h/h) \otimes I, \qquad S_0 \mapsto S_\varepsilon = (1-\varepsilon)S_0 + \varepsilon(I_h/h) \otimes I.$$

Each physical raw Pauli or Pauli-payload observable is traceless. Thus every centered letter becomes exactly $T_\varepsilon = (1-\varepsilon)T_0$, while D stays fixed. Expand $F = \sum_w c_w T_w$ and compute a rational $B_{\deg} \ge \sum_w |w||c_w|$. Telescoping products of contractions gives $\|F(T_\varepsilon) - F(T_0)\| \le \varepsilon B_{\deg}$. A rational

$$0 < \varepsilon < \min\{1, \gamma/(4\max\{1, B_{\deg}\})\}$$

preserves a margin below $-\gamma/4$ with faithful baselines. This choice does not require knowing the witness dimension. We do not claim faithful exact-zero anchor witnesses. □

Substitute (21) into F. The result is a rational Hermitian polynomial p in the actual effects Z_A from (20), with the same positive gap and negative witness. This substitution is finite and purely symbolic.

6 From polynomial positivity to channel order

6.1 Multiplication constraints as inverse Sylvester kernels

The game polynomial has the right sign, but a Fisher test cannot directly evaluate its products. Its available terms are inverse Sylvester kernels. We replace each word by a vector variable and enforce multiplication along a finite graph. The key is a completed square built entirely from the available inverse kernels: its zero set is one multiplication step. On the exact graph the objective is the original polynomial; off the graph a finite penalty is sufficient because the positive branch has a supplied gap.

We give the scalar-coefficient construction, which suffices here. Matrix coefficients require only an untouched coefficient factor. Suppose $\alpha I \le A_j \le (1-\alpha)I$, and let

$$p(Z) = \sum_w c_w Z_w, \qquad Z_{i_1 \cdots i_k} = Z_{i_1} \cdots Z_{i_k}, \qquad c_{\mathrm{rev}(w)} = \overline{c_w}.$$

Given a positive rational gap g, we construct fixed rational Hermitian residues H_0, H_j^+, H_j^- such that

$$K(R,S) = H_0 \otimes D^{-1} + \sum_j (H_j^+ \otimes B_{A_j}^{-1} + H_j^- \otimes B_{I-A_j}^{-1}) \tag{47}$$

has a uniform gap whenever $p(Z) \ge gI$, and has a negative witness whenever p does.

6.1.1 An exact edge penalty

For $B = B_{A_j}$ and $u = D^{1/2}x$, $w = D^{1/2}y$, define

$$Q_j(u,w) = \langle u, B^{-1}u\rangle + \langle w-u, (D-B)^{-1}(w-u)\rangle - \langle w, D^{-1}w\rangle.$$

Completing the square in the single Hermitian operator $Z = D^{-1/2}BD^{-1/2}$ gives

$$Q_j(u,w) = \langle x - Zy, [Z(I-Z)]^{-1}(x-Zy)\rangle, \qquad 4\|x-Zy\|^2 \le Q_j(u,w). \tag{48}$$

The identity uses $Z^{-1} + (I-Z)^{-1} = [Z(I-Z)]^{-1}$ and involves no commutation of B with D. Its zero set is exactly the desired edge recurrence $x = Zy$.

6.1.2 Suffix nodes and explicit residues

For example, the word $Z_1Z_2Z_3$ requires the chain $z_\varnothing \mapsto z_3 \mapsto z_{23} \mapsto z_{123}$, with $z_{jw} = Z_j z_w$. Suffixes, rather than prefixes, put the rightmost factor next to the root and preserve the order of a noncommutative word.

Let $\mathcal{W}$ be the smallest suffix-closed set containing the empty word $\varnothing$ and every supported word of p. Write $N_w = |\mathcal{W}|$ and $L_w = \sum_{v \ne \varnothing} |v|$. Each nonempty node $v = jw$ has parent w and the intended recurrence $z_v = Z_j z_w$. Suffix closure preserves the declared noncommutative word order. Define a Hermitian node matrix H_p by

$$(H_p)_{\varnothing,\varnothing} = c_\varnothing, \qquad (H_p)_{\varnothing,w} = c_w/2, \qquad (H_p)_{w,\varnothing} = \overline{c_w}/2 \quad (w \ne \varnothing).$$

Its quadratic form is

$$P(z) = c_\varnothing \|z_\varnothing\|^2 + \mathrm{Re} \sum_{w \ne \varnothing} c_w \langle z_\varnothing, z_w\rangle.$$

On the exact graph it equals $\langle z_\varnothing, p(Z) z_\varnothing\rangle$. For standard node vectors e_v, and penalty $M_1 \ge 0$, set

$$\begin{aligned} H_0 &= H_p - M_1 \sum_{v=jw} e_w e_w^*, \\ H_j^+ &= M_1 \sum_{v=jw} e_v e_v^*, \qquad H_j^- = M_1 \sum_{v=jw} (e_w - e_v)(e_w - e_v)^*. \end{aligned} \tag{49}$$

With $U_v = D^{1/2} z_v$ the kernel form is exactly

$$\langle U, KU\rangle = P(z) + M_1 \sum_{v=jw} Q_j(U_v, U_w). \tag{50}$$

All output residues are constant. No square root of a baseline is required in the algorithm; square roots occur only in this proof.

Lemma 6.1 (Uniform compiler gap)**.** *Choose rational $c'_w \ge |c_w|$, put $b = \sum_{w\neq\varnothing} |w| c'_w$, and set*

$$M_1 = b^2/(8g) + 1/4, \qquad \eta = \frac{g}{4N_w + 2gL_w}.$$

If $p(Z) \ge gI$ on the actual domain, then

$$K(R,S) \ge \eta I \otimes D^{-1} \tag{51}$$

on that domain. Any strict negative witness for p transfers exactly to K at the same baselines and reference sizes.

Proof. For arbitrary nodes let $a = \|z_\varnothing\|$, $r_{jw} = z_{jw} - Z_j z_w$, and $t^2 = \sum_{v\neq\varnothing} \|r_v\|^2$. Expanding the recurrence along each path, using contractions, gives

$$\|z_v - Z_v z_\varnothing\| \le \sqrt{|v|}\, t, \qquad \sum_v \|z_v - Z_v z_\varnothing\|^2 \le L_w t^2.$$

Since the objective has only root-to-word terms,

$$|P(z) - \langle z_\varnothing, p(Z) z_\varnothing\rangle| \le bat.$$

By (48) and Young's inequality,

$$\langle U, KU\rangle \ge ga^2 - bat + 4M_1 t^2 \ge (g/2)a^2 + t^2.$$

Also $\|z\|^2 \le 2N_w a^2 + 2L_w t^2$, which proves (51) with the displayed conservative η, including the constant-polynomial case. If $\langle \xi, p(Z)\xi\rangle < 0$, set $z_v = Z_v \xi$ and $U_v = D^{1/2} z_v$. Every edge penalty is zero, so (50) equals the negative polynomial value exactly. □

All constants and residues are rational complex when the polynomial and effects are. Bounds such as $c'_w = |\operatorname{Re} c_w| + |\operatorname{Im} c_w|$ suffice. There are at most $2g_0 + 1$ distinct denominators for g_0 input letters, each a fixed strictly positive physical effect, and $N_w \le 1 + \sum_{w\neq\varnothing, c_w\neq 0} |w|$. The positive gap is essential for this finite-penalty argument. Mere nonnegativity of p is not being compiled by the assertion.

6.2 Putting the kernels into measurement channels

To realize a kernel as a Fisher difference, its residues become physical tangent coordinates and its denominators become measurement outcomes. There are two constraints. Signed residues must be represented without introducing irrational spectral coefficients, and Hermitian tangents must retain both real and imaginary parts of a rectangular kernel witness. A rational signed rank-one decomposition solves the first; two conjugate quadratures solve the second.

Start with a finite rational Hermitian kernel

$$K(R,S) = \sum_r H_r \otimes B_{A_r}(R,S)^{-1}, \qquad A_r > 0, \tag{52}$$

and a supplied gap $K \ge \eta I \otimes B_I^{-1}$, or a strict negative witness. Let the residue dimension be d and the denominator physical dimension be p_0. Choose q_0 with $p_0 q_0 \ge d$ and distinct physical matrix units $P_\alpha = |a_\alpha\rangle\langle p_0 + b_\alpha|$ in the upper-to-lower block. For $v \in \mathbb{C}^d$ set $V(v) = \sum_\alpha v_\alpha P_\alpha$.

6.2.1 Rational signed rank one decomposition

Every rational Hermitian H_r has a finite decomposition

$$H_r = \sum_t c_t v_t v_t^*, \qquad c_t \in \mathbb{Q} \setminus \{0\}, \quad v_t \in \mathbb{Q}(\mathrm{i})^d. \tag{53}$$

One elementary construction uses diagonal units and the vectors $e_j \pm e_k$, $e_j \pm \mathrm{i}e_k$: differences of their outer products span the real and imaginary off-diagonal Hermitian matrix units. Thus no irrational spectral square roots are needed.

For every signed term, make *two* baseline outcomes with upper block $\lambda A_r/|c_t|$ and lower block $\lambda \operatorname{Tr}(A_r)/(p_0|c_t|)\, I_{q_0}$. Choose a small positive rational λ so that the sum of all upper blocks is strictly less than I_{p_0}, and add a common filler

$$D_0 = \operatorname{diag}(A_0, [\operatorname{Tr} A_0/p_0]I_{q_0}), \qquad A_0 = I_{p_0} - \sum_{t,\ell=1,2} \lambda A_{r(t)}/|c_t| > 0.$$

The baseline list (D_i) is a strictly positive rational POVM. For term t its two Hermitian off-diagonal numerators are

$$O_{t,1} = V(v_t) + V(v_t)^*, \qquad O_{t,2} = \mathrm{i}V(v_t) - \mathrm{i}V(v_t)^*.$$

Put these numerators on the source side if $c_t > 0$, on the target side if $c_t < 0$, and put zero on the other side. The filler has zero numerator on both sides. Denote the resulting lists by O_i^E, O_i^F.

6.2.2 Effective baselines and both quadratures

For a faithful physically block-diagonal base $\bar{R} = \operatorname{diag}(R_0, T)$, let

$$R_{\mathrm{eff}} = R_0 + (I_{p_0}/p_0) \otimes \operatorname{Tr}_{q_0} T, \qquad W = \operatorname{Tr}_{p_0} R_{\mathrm{eff}} = \operatorname{Tr}_{p_0+q_0} \bar{R}. \tag{54}$$

The output of D_i is exactly the corresponding scaled $\mathcal{A}_{A_r}(R_{\mathrm{eff}})$. Conversely every $R_{\mathrm{eff}} > 0$ is realizable with both blocks faithful by taking $T = tI$ for sufficiently small $t > 0$ and subtracting its contribution from R_{eff}.

Write a physical cross tangent as

$$X = \sum_{\alpha=1}^{d} (P_\alpha \otimes Z_\alpha + P_\alpha^* \otimes Z_\alpha^*).$$

For one term let $U_t = \sum_\alpha \overline{(v_t)_\alpha} Z_\alpha$. Its two outputs are $U_t + U_t^*$ and $\mathrm{i}U_t^* - \mathrm{i}U_t$, up to an immaterial overall sign for the second. Because Ω^{-1} respects adjoints, their squared norms sum to $4\langle U_t, \Omega^{-1}(U_t)\rangle$. In particular the two quadratures eliminate extra conjugate cross terms for arbitrary tangents, not just for one-sided rectangular slices. Define

$$\Gamma_{\bar{R}}(X) = \sum_i \left[g_{\bar{R}_i}(\operatorname{Tr}_H(O_i^E X)) - g_{\bar{R}_i}(\operatorname{Tr}_H(O_i^F X)) \right], \quad \bar{R}_i = \operatorname{Tr}_H(D_i \bar{R}). \tag{55}$$

The precise normalization is

$$\Gamma_{\bar{R}}(X) = \frac{8}{\lambda} \langle \operatorname{vec} Z, K(R_{\mathrm{eff}}, R_{\mathrm{eff}}) \operatorname{vec} Z \rangle. \tag{56}$$

The factor eight is 4 from the quadratures and 2 from $\Omega^{-1} = 2\mathcal{J}^{-1}$; the coefficient $|c_t|/\lambda$ is absorbed by the baseline scaling.

Let $\mathcal{C}$ be the real span of the off-diagonal numerator matrices. Give its coefficients the Hilbert–Schmidt orthonormal physical-coordinate norm

$$|X|_W^2 = \sum_\mu g_W(X_\mu), \qquad X = \sum_\mu C_\mu \otimes X_\mu.$$

On the full selected complex-coordinate space,

$$|X|_W^2 = 4\sum_\alpha \langle Z_\alpha, \mathcal{J}_{W,W}^{-1}(Z_\alpha)\rangle. \tag{57}$$

The same identity restricts to $\mathcal{C}$. Any physical cross direction orthogonal to $\mathcal{C}$ is invisible to both signal lists and will require no coercivity. From (56) and the kernel gap we get the exact cross gap

$$\Gamma_{\bar{R}}(X) \geq \gamma_{\mathrm{cross}}|X|_W^2, \qquad \boxed{\gamma_{\mathrm{cross}} = 2\eta/\lambda}. \tag{58}$$

For a negative rectangular kernel witness at R, S, use effective baseline $R \oplus S$ on a direct-sum reference and put each Z_α only in its R-to-S reference block. The square-matrix form restricts exactly to the rectangular Sylvester form. Formula (56) is strictly negative. Realize $R \oplus S$ by faithful physical lower padding using (54). If necessary project the tangent onto $\mathcal{C}$; its signal outputs and negative value are unchanged. This constructs a faithful physical-block-diagonal negative slice witness.

6.3 Completing the measurement comparison globally

So far the desired comparison is exact only at block-diagonal bases and cross-block tangents. Complete order asks for all bases and all tangents. Two additions close this gap. Pairing each signal outcome with its negative cancels odd perturbation terms and leaves an exact resolvent square. Additional block-diagonal measurements then dominate the remaining tangent directions.

There is one unavoidable null direction: a perturbation of the untouched reference is seen equally by every trace-preserving channel. We first subtract that common reference score. Only after taking this quotient can the additional measurements supply a coercive bound. The original negative witness lies entirely in the cross-block slice, so these additions leave its sign unchanged.

6.3.1 Signal pairs and informationally complete padding

Given (D_i) and O_i^E, O_i^F above, define signal POVMs

$$E_{i,\pm} = (D_i \pm \delta O_i^E)/2, \qquad F_{i,\pm} = (D_i \pm \delta O_i^F)/2.$$

Their sums are I. For sufficiently small rational $\delta > 0$, every effect is strictly positive. Choose a strictly positive rational block-diagonal informationally complete POVM (G_h) whose real span is the entire physical block-diagonal Hermitian subspace. For example, for a rational Hermitian basis (B_μ) choose sufficiently small rational $t > 0$ and use effects $(I \pm tB_\mu)/(2L)$, where L is the number of basis elements. This gives exact normalization and spanning.

If $(\widetilde{B}_\mu)$ is an orthonormal basis of that subspace, let

$$g_{h\mu} = \mathrm{Tr}(G_h \widetilde{B}_\mu), \qquad c > 0 \quad \text{be a certified lower bound for } \lambda_{\min}\Big(\sum_h g_h g_h^{\mathsf{T}}\Big).$$

Such a positive rational lower bound is effectively computable by finite algebraic estimates. The basis used for estimates need not be rational; all physical effects are rational. The final source POVM consists of $E_{i,\pm}/2$ and $G_h/2$. The final target consists of $F_{i,\pm}/2$ and a single erasure effect $I/2$. They are normalized strictly positive rational POVMs once δ is chosen.

6.3.2 A reference score gauge

The following subtraction is the SLD analogue of removing a nuisance score common to two statistical experiments. For a faithful physical-reference base R, let $W = \mathrm{Tr}_H R$. For a Hermitian tangent X, put

$$A = \Omega_W^{-1}(\mathrm{Tr}_H X), \qquad X' = X - \Omega_R(I_H \otimes A).$$

Then $\mathrm{Tr}_H X' = 0$. Every physical trace-preserving channel Φ obeys $\Phi_R(\Omega_R(I \otimes A)) = \Omega_{\Phi_R(R)}(I \otimes A)$. Expansion of the inverse-metric square shows

$$g_{\Phi_R(R)}(\Phi_R(X')) - g_{\Phi_R(R)}(\Phi_R(X)) = -2\,\mathrm{Tr}[X(I \otimes A)] + \mathrm{Tr}[R(I \otimes A^2)],$$

independently of Φ. Every source-minus-target difference is therefore unchanged by this gauge.

Decompose $X' = Y + X_{\mathcal{C}} + X_{\mathrm{inv}}$ into physical diagonal, relevant cross, and invisible cross parts. The erasure branch sees zero. The source padding sees only Y and contributes at least

$$\sum_h g_{\mathrm{Tr}_H(G_h R)}(\mathrm{Tr}_H(G_h Y)) \geq c|Y|_W^2. \tag{59}$$

Indeed $G_h \leq I$ makes its base at most W, so inverse Jordan ordering bounds each term below by its W-norm; the frame lower bound gives the last inequality. The gauge is essential: no false coercivity assertion is made on reference-only tangents before this quotient.

6.3.3 An exact paired resolvent identity

For one signal outcome on either side abbreviate

$$R_i = \mathrm{Tr}_H(D_i R), \quad V = \mathrm{Tr}_H(OR), \quad U = \mathrm{Tr}_H(D_i Y), \quad Z = \mathrm{Tr}_H(OX_{\mathcal{C}}),$$

$$J = \Omega_{R_i}, \quad K = \Omega_V, \quad u = J^{-1/2}U, \quad z = J^{-1/2}Z, \quad T = J^{-1/2}KJ^{-1/2}.$$

The sum of the two sign-outcome Fisher costs is exactly

$$F_{\mathrm{pair}} = \|u\|^2 + \delta^2\langle z - Tu, (I - \delta^2 T^2)^{-1}(z - Tu)\rangle. \tag{60}$$

To verify it, write the pair as one half the sum of the inverse quadratic forms of $(u \pm \delta z, I \pm \delta T)$. The even inverse average is $(I - \delta^2 T^2)^{-1}$ and the odd difference is $-\delta T(I - \delta^2 T^2)^{-1}$. Expanding proves the completed square. Every odd order cancels. The base term $\|u\|^2$ is identical on source and target and cancels in their difference.

6.3.4 Controlling all tangent directions

Let $\bar{R}$ be the physical block pinching of R, which leaves R_i and W unchanged. Put $a = |X_{\mathcal{C}}|_W$, $b = |Y|_W$ and $\gamma = \gamma_{\mathrm{cross}}$. The paired identity has leading term $\delta^2\Gamma_{\bar{R}}(X_{\mathcal{C}})$, with its known positive cross gap. Its remaining mixed and resolvent terms admit bounds in these same two norms, uniformly in the reference dimension and the conditioning of R. Appendix C gives explicit rational constants $C_0 > 0$ and smallness conditions on δ for which

$$\Delta_{\mathrm{signal}} \geq \delta^2[\gamma a^2/4 - C_0 b^2].$$

Here Δ_{signal} is the source-minus-target Fisher difference of the signal POVMs. Informationally complete padding supplies the missing cb^2. Since signal and padding have equal final weights,

$$2\Delta_{\mathrm{final}} \geq \frac{\delta^2\gamma}{4}a^2 + (c - \delta^2 C_0)b^2. \tag{61}$$

Choose a positive dyadic δ satisfying those smallness conditions and $\delta^2 C_0 \le c$. The same conditions make every effect strictly positive. All choices depend only on the finite physical effect lists. Thus the cross gap absorbs mixed errors, while the additional measurements absorb the remaining diagonal loss.

Proposition 6.2 (Global completion)**.** *The preceding finite rational construction converts the supplied kernel gap into complete SLD dominance of the final source POVM over the target POVM. A negative kernel witness converts into a genuine finite faithful SLD violation.*

Proof. The positive assertion was proved for every faithful base and every Hermitian tangent after an invariant gauge, so it is complete dominance. For the negative assertion use the physically block-diagonal base and pure cross tangent of Section 6.2. Then $V = 0$ and $U = 0$ in every pair, and both padding tangents vanish. The final difference is exactly

$$\frac{\delta^2}{2}\Gamma_R(X) < 0.$$

It is an exact metric witness with no perturbative error. If the base is unnormalized, divide both it and the tangent by its trace; the difference is divided by the same positive scalar. The tangent is traceless, and the base is faithful. □

7 Decidability at fixed size and undecidability in general

Apply the game reduction with $r = 3/4$ and $\gamma = 1/4$. In the nonhalting case the external game has $\omega^* \le 1/2 = r - \gamma$, so Proposition 5.3 gives a uniform polynomial gap $g = \gamma/2 = 1/8$. Lemma 6.1 yields a positive inverse-kernel gap, and Proposition 6.2 yields complete SLD dominance.

In the halting case the game has a finite perfect strategy; in particular it has a finite strategy scoring above $7/8 = r + \gamma/2$. The faithful polynomial witness transfers exactly through the graph compiler and the negative POVM slice. Complete SLD dominance fails. Every output operation uses finite rational arithmetic and effective bounds; no witness dimension is needed to compute the POVMs. We have therefore reduced halting to failure and nonhalting to dominance.

7.1 Failure is recursively enumerable

Enumerate reference dimensions and rational complex faithful density matrices ρ together with rational complex traceless Hermitian tangents H. Exact rational linear algebra decides Hermiticity, normalization and strict positive definiteness. For each strictly positive rational POVM and faithful rational input the output blocks are positive definite and rational. Solve each rational Sylvester system

$$\Omega_{\rho_i^E}(L_i) = H_i^E$$

exactly. Its Fisher value is rational, so the source-minus-target sign is decidable. Every strict real violation persists on an open neighborhood by continuity of inversion on positive definite matrices. Rational faithful density matrices and rational traceless Hermitian tangents are dense in the required affine spaces. Thus the enumeration eventually detects every failure. It may run forever precisely on dominating pairs.

The same upper bound holds for arbitrary rational-complex Choi channel pairs. Their faithful rational inputs have rational output bases and tangents. On the full output space, solve the possibly singular rational linear system $\Omega_A(L) = K$. A solution exists because A is faithful on the fixed channel support and K is supported there. Rational Gaussian elimination

supplies a rational solution; its value $\operatorname{Tr} KL$ is independent of the unconstrained kernel block. Continuity on the fixed effective supports again makes rational witnesses complete. Strictly positive measurement pairs already give hardness for this larger class.

Failure is in RE and is RE-hard by the halting reduction; hence it is RE-complete. The dominating set is coRE-complete. By Section 3, exactly the same set of pairs defines the unrestricted entropy order, proving the order classifications of Theorem 2.2.

7.2 No computable sufficient SLD reference cutoff

For a fixed reference dimension n, existence of a faithful SLD violation is a first-order real-algebraic question. Use real and imaginary matrix coordinates for ρ, H, L_i; impose trace and Hermiticity constraints, positive definiteness by principal-minor conditions, and the polynomial Sylvester equations. The Fisher difference is polynomial in these variables once the scores L_i are included. Real quantifier elimination therefore decides whether a violation exists at that fixed size.

Suppose a total computable bound $b(E, F)$ were sufficient to detect every failure. Compute it and decide the finitely many reference sizes $1, \ldots, b(E, F)$ by quantifier elimination. Absence of a violation would certify complete dominance, while its presence would certify failure. This contradicts undecidability and proves the reference-cutoff assertion of Theorem 2.2.

This argument specifically establishes the reference-cutoff obstruction for SLD tests. The entropy decision problem is equally undecidable by order equivalence; no fixed-dimension algorithm for transcendental entropy formulas is assumed in the cutoff argument.

8 A bounded defect and fixed-error inapproximability

Exact-order undecidability does not by itself rule out approximation: the violating entropy difference could be too small to resolve. To prove Theorem 2.3, we first bound the defect $\mathcal{V}$ from (5) and prove its additivity. We then obtain a positive margin computable from the game, despite the unknown witness dimension. Tensor powers amplify that margin to any fixed numerical scale.

8.1 A bounded, additive information defect

Although each output divergence can contain arbitrarily much reference information, both measurements retain exactly the same reference. Subtracting its divergence exposes the bounded contribution of the measurement outcome.

Lemma 8.1 (Boundedness of the defect)**.** *If $a_j I \le F_j \le b_j I$ with $a_j > 0$, then*

$$0 \le \mathcal{V}(E, F) \le \max_j \log(b_j/a_j) < \infty.$$

Proof. For any strictly positive POVM $M = (M_i)$, choose $\alpha_i I \le M_i \le \beta_i I$ with $\alpha_i > 0$. Let ρ_R, σ_R be the reference marginals and ρ_i, σ_i the measurement output blocks. Positivity of partial trace gives $\alpha_i \rho_R \le \rho_i \le \beta_i \rho_R$, and similarly for σ_i. Operator monotonicity of logarithm implies

$$(\log \alpha_i) I \le \log \rho_i - \log \rho_R \le (\log \beta_i) I,$$

with the same bounds for $\log \sigma_i - \log \sigma_R$. Since $\sum_i \rho_i = \rho_R$, the reference contribution cancels

exactly in

$$G_M(\rho,\sigma) := D(\mathcal{M}_R(\rho)\|\mathcal{M}_R(\sigma)) - D(\rho_R\|\sigma_R)$$
$$= \sum_i \operatorname{Tr}\rho_i\big[(\log\rho_i - \log\rho_R) - (\log\sigma_i - \log\sigma_R)\big].$$

Data processing under forgetting the outcome gives $G_M \geq 0$; the displayed logarithm bounds give $G_M \leq \max_i \log(\beta_i/\alpha_i)$. The defect in (5) equals $G_F - G_E$, yielding the upper bound. Taking $\rho = \sigma$ proves nonnegativity of the supremum. □

Proposition 8.2 (Additivity and effective lower bounds)**.** *For strictly positive POVM pairs,*

$$\mathcal{V}(E_1 \otimes E_2, F_1 \otimes F_2) = \mathcal{V}(E_1, F_1) + \mathcal{V}(E_2, F_2).$$

The value $\mathcal{V}(E,F)$ is lower semicomputable on rational-complex strictly positive POVMs: an algorithm enumerates rational lower bounds converging to it. Its strict-positivity set is RE*-complete.*

Proof. For the upper bound, insert $\mathcal{F}_1 \otimes \mathcal{E}_2$ between $\mathcal{E}_1 \otimes \mathcal{E}_2$ and $\mathcal{F}_1 \otimes \mathcal{F}_2$. Each increment is a one-factor defect with all other inputs and outputs treated as an untouched reference, hence is bounded by the corresponding $\mathcal{V}$. Faithful input pairs remain faithful block-output pairs under a strictly positive measurement: an effect $M_j \geq a_j I$ gives an output block at least a_j times the faithful reduced input. Thus both one-factor bounds apply. Product faithful witnesses and relative-entropy additivity give the reverse inequality, by approaching the two suprema.

For lower semicomputability, enumerate reference dimensions and rational faithful state pairs. Their output entropies and relative entropies are computable in fixed dimension; dovetailed certified lower bounds and running maxima approach the supremum by density and continuity. Finally $\mathcal{V} > 0$ is exactly failure of complete entropy order: a strict boundary violation can be made faithful by a common regularization, or one can use the saturated construction from a finite faithful SLD violation. Theorem 2.2 gives RE-completeness. □

Boundedness alone does not provide a computable approximation procedure. To rule one out we need more than the strict witnesses used in the exact-order reduction: their absolute entropy margin must have a positive lower bound computable from the game, without knowing the strategy dimension. The following two sections establish this extra normalization.

8.2 A witness margin independent of the strategy dimension

The exact-order reduction can tolerate an arbitrarily small violation. Approximation hardness cannot: it needs a positive entropy margin computable from the game description. Simply normalizing a large witness may destroy such a margin. We avoid that loss by choosing a maximally entangled strategy sector. Its word matrices have operator norm of order $\ell^{-1/2}$, where ℓ is the unknown local dimension. Normalizing the baseline by ℓ and the tangent by $\sqrt{\ell}$ then cancels the dimension in the Fisher quadratic form.

Here we use the finite perfect PCC strategy supplied by [1, Theorem 12.6, p. 172; Definitions 5.9–5.11, pp. 51–52]. Recall its consistency condition: every answer projector M in either family satisfies

$$(M \otimes I)\psi = (I \otimes M)\psi. \tag{62}$$

The additional question-pair commutation condition in PCC is not needed below.

Lemma 8.3 (A perfect maximally entangled sector)**.** *A finite perfect strategy with the consistency property* (62) *has a finite perfect maximally entangled strategy obtained by restriction to a Schmidt sector.*

Proof. Write $\psi = \operatorname{vec} C$, with $\|C\|_{\mathrm{HS}} = 1$. Consistency says $MC = CM^{\mathsf{T}}$. Since M is Hermitian, its adjoint relation gives

$$MCC^* = CM^{\mathsf{T}}C^* = CC^*M.$$

Thus every Alice answer projector preserves the spectral sectors of her reduced state CC^*. The same reasoning for the other family proves that every Bob answer projector preserves the matching sectors of $(C^*C)^{\mathsf{T}}$. For each positive Schmidt eigenvalue λ of multiplicity d_λ, the state decomposes as

$$\psi = \sum_\lambda \sqrt{\lambda d_\lambda}\, \psi_\lambda, \qquad \sum_\lambda \lambda d_\lambda = 1,$$

where ψ_λ is a unit maximally entangled vector between the matching d_λ-dimensional sectors. Sector preservation and orthogonality make every game probability the corresponding convex combination of sector probabilities. Since payoffs lie in $[0,1]$, a score of one forces each nonzero sector to score one. Restrict both measurement families to any such sector. □

This is a property of the hard game family. No maximally entangled optimality assertion for arbitrary games is used.

8.2.1 Faithful Choi regularization with a fixed marginal

Keep $r = 3/4$, $\gamma = 1/4$ in the reduction. Let $a = 2MK$ be the initial physical dimension and let $L = N_w$ be the compiler node count. Use lower physical dimension $q_0 = \lceil L/a \rceil$ in Section 6.2. Write $\lambda > 0$ for its baseline scaling and $\delta_s > 0$ for the signal parameter of Section 6.3. All are computed from the game.

Choose rational $B_{\deg} \geq \sum_w |w||c_w|$ for the centered-letter polynomial $F = \sum_w c_w T_w$, and put

$$e = \min\{1/2, \gamma/(8\max\{1, B_{\deg}\})\}. \tag{63}$$

A perfect maximally entangled strategy, together with the Bell anchor, has local dimension ℓ and root $z = \operatorname{vec}(I_\ell)/\sqrt{\ell}$ after choosing local bases. Its Choi baselines have reference marginal I_ℓ. Regularize them by

$$R_e = (1-e)R_0 + e(I_a/a) \otimes I_\ell, \qquad S_e = (1-e)S_0 + e(I_a/a) \otimes I_\ell.$$

Their marginals stay fixed and

$$(e/a)I \leq R_e, S_e \leq aI. \tag{64}$$

For the upper bound, any positive block matrix with physical partial trace I obeys $R \leq a\sum_i |i\rangle\langle i| \otimes R_{ii} \leq aI$ by the pinching inequality and $\sum_i R_{ii} = I$. The perfect honest polynomial expectation is $r - 1 = -\gamma$. Every centered letter is multiplied by $1 - e$, so

$$\langle z, F(T_e)z\rangle \leq -\gamma + eB_{\deg} \leq -\gamma/2. \tag{65}$$

8.2.2 Operator norm control of word witnesses

At these marginal-identity baselines, $D = 2I$. Upon reshaping, each normalized positive-effect letter acts as

$$Z \longmapsto (C_A Z + ZD_A)/2, \qquad 0 \leq C_A, D_A \leq I,$$

up to the fixed vectorization transpose. It therefore contracts operator norm. Starting from $I_\ell/\sqrt{\ell}$, every word-node matrix has operator norm at most $1/\sqrt{\ell}$. The exact graph coordinates $U_w = \sqrt{2}z_w$ have reshaped norms at most $\sqrt{2}/\sqrt{\ell}$.

On the reference $\mathbb{C}^\ell \oplus \mathbb{C}^\ell$, put these matrices in the upper-right reference block of the L selected physical cross coordinates. Let Z be the resulting rectangular physical block and $X = \left(\begin{smallmatrix} 0 & Z \\ Z^* & 0 \end{smallmatrix}\right)$ the Hermitian physical-cross tangent. Triangle inequality gives

$$\operatorname{Tr} X = 0, \qquad \|X\| \le 2L/\sqrt{\ell}. \tag{66}$$

The operator norm, rather than only a Hilbert–Schmidt bound, is the reason maximally entangled completeness is useful here.

8.2.3 Physical padding and a dimension independent Fisher gap

Let $R_{\mathrm{eff}} = R_e \oplus S_e$, and define the physical-block-diagonal baseline

$$T_{\mathrm{low}} = \frac{e}{2q_0} I_{q_0} \otimes I_{2\ell}, \qquad R_{\mathrm{top}} = R_{\mathrm{eff}} - \frac{e}{2a} I_a \otimes I_{2\ell}, \qquad \Theta_0 = \operatorname{diag}(R_{\mathrm{top}}, T_{\mathrm{low}}).$$

Its effective baseline (54) is exactly R_{eff}, its reference marginal is $I_{2\ell}$, and $\operatorname{Tr}\Theta_0 = 2\ell$. Moreover

$$\frac{e}{2\max\{a, q_0\}} I \le \Theta_0 \le aI.$$

Normalize by

$$\rho = \Theta_0/(2\ell), \qquad H = X/\sqrt{\ell}, \qquad h = \frac{8L\max\{a, q_0\}}{e}. \tag{67}$$

Equations (66)–(67) imply

$$\frac{e}{4\max\{a, q_0\}\ell} I \le \rho \le \frac{a}{2\ell} I, \qquad \|H\| \le \frac{2L}{\ell}, \qquad -h\rho \le H \le h\rho.$$

Let $\alpha > 0$ be a certified rational lower bound for every final effect on both sides. For either final channel Φ, set $A_\Phi = \Phi_R(\rho)$ and $K_\Phi = \Phi_R(H)$. Since $\rho_R = I_{2\ell}/(2\ell)$, every outcome block obeys

$$\frac{\alpha}{2\ell} I \le (A_\Phi)_j \le \frac{1}{2\ell} I.$$

Thus each full output state has condition number at most $\kappa = 1/\alpha$, independently of ℓ. Positivity also preserves $-hA_\Phi \le K_\Phi \le hA_\Phi$.

For the unnormalized base and cross tangent, the exact graph identity and (56) give

$$\Gamma_{\Theta_0}(X) = \frac{8}{\lambda} \langle z, F(T_e) z \rangle.$$

The final signal/padding branch weight is $1/2$ and its signal factor is δ_s^2. Hence the source-minus-target Fisher difference is at most $-2\delta_s^2\gamma/\lambda$ on (Θ_0, X). The change to (ρ, H) multiplies this difference by 2, because the metric is inverse-linear in its base and quadratic in its tangent. Consequently

$$g_{A_{\mathcal{F}}}(K_{\mathcal{F}}) - g_{A_{\mathcal{E}}}(K_{\mathcal{E}}) \ge \delta_G, \qquad \delta_G = 4\delta_s^2\gamma/\lambda > 0. \tag{68}$$

All three constants h, κ, δ_G are positive rational bounds computable solely from the game description. Neither the faithful normalization nor the absolute Fisher gap loses an unknown factor of the strategy dimension.

8.3 Converting a Fisher gap to an effective entropy gap

A positive Fisher gap is still an infinitesimal statement. The reference filter of Section 3 must now be chosen at a finite resolution, and its perturbation at a finite size. Two bounds make that choice effective: an output condition-number bound controls the filter error, and a relative tangent bound controls the logarithm's Taylor remainder. Both bounds above are independent of the strategy dimension.

Lemma 8.4 (Uniform entropy witness)**.** *Suppose a faithful input state ρ and Hermitian H satisfy $-h\rho \le H \le h\rho$ for rational $h > 0$. Suppose both effective output states of channels $\mathcal{P}, \mathcal{Q}$ have condition number at most rational $\kappa \ge 1$, and their target-minus-source SLD gap is at least rational $d_0 > 0$. There is a computable rational $\varepsilon_0 > 0$ and a finite auxiliary reference factor depending only on h, κ, d_0, for which faithful input state pairs attain target-minus-source Umegaki gap at least ε_0.*

The proof chooses a narrow enough finite filter to retain half the Fisher gap, adds a small faithful component to the numerator, and then chooses a perturbation for which the quadratic gain dominates the cubic remainder. Appendix B gives all rational choices and estimates. The auxiliary dimension and the positive margin depend only on h, κ, d_0, not on the original reference dimension. They need not be efficient bounds. As throughout, the two reference marginals may differ.

8.4 Approximation remains undecidable

Theorem 8.5 (No total additive approximation algorithm)**.** *There is no total algorithm which, given rational complex strictly positive POVMs E, F and rational $\varepsilon > 0$, returns a rational v satisfying*

$$|v - \mathcal{V}(E, F)| \le \varepsilon.$$

The conclusion holds with the supremum in (5) *restricted to faithful state pairs, for which the value is finite and nonnegative.*

Proof. Given a Turing machine, compute its hard game and the POVMs from the preceding reduction. In the nonhalting case complete SLD and therefore complete entropy dominance hold, so $\mathcal{V} = 0$. In the halting case Lemma 8.3 gives a perfect maximally entangled strategy. Section 8.2 supplies a faithful Fisher witness and game-computable rational constants h, κ, δ_G satisfying Lemma 8.4. Apply that lemma with $d_0 = \delta_G$ to compute a positive rational ε_G and a faithful entropy witness with defect at least ε_G. All constants can be computed without finding the strategy or its dimension. Thus halting implies $\mathcal{V} \ge \varepsilon_G$.

A purported approximation algorithm at tolerance $\varepsilon_G/4$ would return $v \le \varepsilon_G/4$ in the nonhalting case and $v \ge 3\varepsilon_G/4$ in the halting case. Comparing with $\varepsilon_G/2$ would decide halting, a contradiction. □

The primitive gap ε_G depends effectively on the game. This is enough to amplify to a universal numerical accuracy obstruction.

Proof of Theorem 2.3. For the hard pair compute the primitive margin $\varepsilon_G > 0$ and choose $k = \lceil 4c/\varepsilon_G \rceil$. Tensor-product effects are rational, strictly positive and normalized. By Proposition 8.2, the nonhalting branch has defect zero and the halting branch has defect at least $k\varepsilon_G \ge 4c$. An additive-c approximation lies at most at c in the first case and at least at $3c$ in the second. Thresholding at $2c$ decides halting. □

Tensor amplification is a computable reduction, without an output-size bound. Its effective primitive margin is essential: qualitative order undecidability alone would not prove the fixed-error result.

A bare approximate reference-dimension cutoff, without an effective fixed-reference optimization routine, is a separate question. Fixed-reference faithful state pairs can still be arbitrarily ill conditioned. The argument above therefore makes no unqualified approximate-reference-cutoff claim. It does exclude a cutoff together with a total fixed-reference approximation procedure sufficient to approximate $\mathcal{V}$ at every requested accuracy.

9 Scope and further questions

The comparison studied here allows every finite quantum reference and every pair of states, with no equality constraint on their reference marginals. It is weaker than requiring a physical degrading map; comparison by guessing probability or min-entropy has a different relation to degradation [12, 13]. The complement reversal is a statement about these complete information orders, not about replacing a genuine environment by an algebraically convenient channel.

The hardness construction uses generic complex effects and opposite Pauli chirality. It does not automatically apply to real or transpose-compatible subclasses, fixed input dimension, or numerical contraction coefficients. Its inverse kernels act on a spatial two-factor domain. A compression mixing those factors can destroy that domain, so general free-spectrahedral results [14] do not supply a reference cutoff without a domain-preservation argument.

The companion capacity manuscript [15] uses the structural equivalences and the rational halting reduction to construct one memoryless channel whose ordinary private and quantum capacities are positive exactly when the order fails. That positivity compiler provides no fixed positive capacity gap. The approximation obstruction proved here concerns $\mathcal{V}$, and does not establish additive inapproximability of those capacities.

A The anchor and smoothing estimates

These estimates justify the local rounding in Section 5. The first recovers a balanced Bell sector without inverting a small marginal eigenvalue. The second shows that the local averaging contraction filters precisely the same energy.

A.1 Conditioning-independent anchor rigidity

Proof of Lemma 5.1. Write $L_i = px_ip$, $R_i = qy_iq$, and $U_i = L_i - R_i$. The column $C : v \mapsto (pv, qv)$ is an isometry, and $T_i = C^*\pi_i C$ for $\pi_i = x_i \oplus y_i$. The residual $\rho_i = \pi_i Cv - CT_iv$ has norm squared $e_i(v)$. Opposite chirality gives, for cyclic (i, j, k),

$$\|[T_i, T_j]v + 2\mathrm{i}U_kv\| \le \sqrt{e_i(v)} + \sqrt{e_j(v)}. \tag{69}$$

Indeed compare T_iT_jv with $C^*\pi_i\pi_jCv$ and reverse i, j. Hence

$$\sum_k \|U_kv\|^2 \le \tfrac{3}{4}c(v) + \tfrac{3}{2}e(v). \tag{70}$$

The same comparison using the anticommutator gives $\|\{T_i, T_j\}v\| \le \sqrt{e_i(v)} + \sqrt{e_j(v)}$ for $i \ne j$.

The conditioning-free step is an inverse-root identity. Put

$$A = r^{-1}, \quad B = s^{-1}, \quad F = A+B, \quad a = A^{1/2}, \quad b = B^{1/2}, \quad \xi = F^{-1/2}v, \quad \alpha = a\xi, \quad \beta = b\xi.$$

All functions of A, B commute, while $[b, x_i] = [a, y_i] = 0$. The orthogonal complement column to C is $(q, -p)$, so

$$e_i(v) = \|d_i\|^2, \quad d_i = F^{-1/2}ab(x_i - y_i)\xi, \quad U_i v = F^{-1/2}(Bx_i - Ay_i)\xi.$$

Adding these formulas gives the exact identity

$$x_i\beta - y_i\alpha = F^{1/2}(a+b)^{-1}(U_i v + d_i). \tag{71}$$

Its multiplier has norm at most one because $\sqrt{a^2+b^2}/(a+b) \le 1$ in commuting functional calculus. Therefore

$$E_* := \sum_i \|x_i\beta - y_i\alpha\|^2 \le 2\sum_i (\|U_i v\|^2 + e_i(v)) \le 5\delta(v).$$

The commuting involutions $c_i = x_i y_i$ satisfy $\sum_i c_i = 4P - I$. Decomposing into P, Q gives

$$E_* = 3\|P(\alpha-\beta)\|^2 + 3\|Q\alpha\|^2 + 3\|Q\beta\|^2 + 2\,\mathrm{Re}\langle Q\alpha, Q\beta\rangle.$$

It follows that

$$\|\alpha-\beta\|^2 \le E_*, \qquad \|Q\alpha\|^2 + \|Q\beta\|^2 \le E_*/2.$$

The scalar inequality $|\sqrt{a^2+b^2} - (a+b)/\sqrt{2}| \le |a-b|/\sqrt{2}$ shows $\|v - (\alpha+\beta)/\sqrt{2}\| \le \|\alpha-\beta\|/\sqrt{2}$. Thus $\|Qv\| \le \sqrt{2E_*}$. Moreover

$$jv = (b-a)(b+a)F^{-1/2}\xi, \qquad \|jv\| \le \sqrt{2}\|\alpha-\beta\| \le \sqrt{2E_*}.$$

This proves the first two bounds in (26); the others use $|\sqrt{(1+t)/2} - 1/\sqrt{2}| \le |t|/\sqrt{2}$ for $-1 \le t \le 1$.

For the action bounds write the components of ρ_i as $a_i' = x_i p v - p T_i v$, $b_i' = y_i q v - q T_i v$. Then

$$U_i v = jT_i v + pa_i' - qb_i', \qquad \|jT_i v\| \le \|U_i v\| + \sqrt{e_i(v)}.$$

For cyclic (i, j, k), the individual commutator residual gives

$$\begin{aligned}\|U_i v\| + \sqrt{e_i(v)} &\le \tfrac{1}{2}\|[T_j, T_k]v\| + \tfrac{1}{2}\sqrt{e_j(v)} + \tfrac{1}{2}\sqrt{e_k(v)} + \sqrt{e_i(v)} \\ &\le \tfrac{\sqrt{7}}{2}\sqrt{\delta(v)} \le 2\sqrt{\delta(v)}.\end{aligned}$$

Expanding $T_i v - x_i v$ with these residuals and the preceding scalar weight estimates gives

$$\|T_i v - x_i v\| \le \|U_i v\| + \|jv\| + (1+\sqrt{2})\sqrt{e_i(v)} \le 7\sqrt{\delta(v)}.$$

Finally let J_H, J_K be fixed local Stinespring isometries for the complete physical algebra. On the enlarged left space extend the weight to $W_\varepsilon = J_H W J_H^* + \varepsilon(I - J_H J_H^*)$, and likewise on the right. For every $\varepsilon > 0$ the exact-representation proof applies. On the embedded original tensor product, D_ε restricts to the original D, and the compression of each normalized letter is the original letter. Off-subspace entries applied to an original vector tend to zero: they contain a factor $\sqrt{\varepsilon}$, while the other original marginal is bounded below for the fixed instance. Uniform boundedness of the letters gives convergence of every fixed finite word. Their energies converge, the local isometries and P remain fixed, and the balance operator on the original subspace is unchanged. Taking $\varepsilon \downarrow 0$ proves the claimed UCP version. No uniform convergence rate is required. □

A.2 Comparison with the smoothing energy

Proof of Lemma 5.2. By Lemma 5.1, the Q and j terms cost at most $10\delta(v)$ each, and $\|jx_iv\| \le \|jT_iv\| + \|T_iv - x_iv\| \le 9\sqrt{\delta(v)}$. Thus $E_0 \le 263\delta$, yielding the first bound.

For the reverse bound put $h_i = \|jx_iv\|$, $h_0 = \|jv\|$, and $b = \|Qv\|$. The exact compression defect is $d_i = qx_ipv - py_iqv$, with $\|d_i\|^2 = e_i(v)$. Replacing p, q by $1/\sqrt{2}$ using the scalar weight estimates, using $\|(x_i - y_i)v\| \le 2b$ and the cross-party commutations, gives

$$\|d_i\|,\ \|U_iv\| \le \sqrt{2}h_0 + h_i + 2b \le \sqrt{7E_0(v)}.$$

Equation (69) gives $\|[T_i, T_j]v\| \le 4\sqrt{7E_0(v)}$. Hence $\delta(v) \le 357E_0(v) \le 1785E_M(v)$, and the stated constant 3000 is sufficient. Finally the scalar inequalities $t^{2n}(1-t) \le 1/(2n+1)$ and $(1-t^n)^2 \le n(1-t)$ on $[0,1]$, applied to M_0, prove (37). □

B An effective finite entropy margin

The qualitative filter limit proves equivalence of orders. For approximation hardness we need a finite filter and a finite perturbation whose errors are controlled without a dimension bound. We give explicit, conservative rational choices; only their computability and positivity are used in the reduction.

Proof of Lemma 8.4. Choose the integer

$$m = \max\{6, \lceil 288\kappa^2h^4/d_0^2\rceil\}, \qquad N = 2m+1, \qquad r_0 = N+1. \tag{72}$$

Use the poles $\lambda_j = 1 + j/m^3$ and partial-fraction coefficients c_j of Lemma 3.2. Define positive rational constants

$$\begin{aligned} \Lambda_0 &= \sum_{j=0}^{2m} |c_j|, & q_* &= h\kappa\Lambda_0, & j_* &= 5^{-(4m+2)}/N, \\ \eta_0 &= \min\{1/4, j_*d_0/(16q_*^2)\}, & & & & \\ t_0 &= \min\{1/(2q_*), 3j_*d_0/(256q_*^3)\}, & \varepsilon_0 &= j_*d_0t_0^2/16. & & \end{aligned} \tag{73}$$

On an auxiliary centre and N leaves take

$$W = \operatorname{diag}(1, \lambda_0, \dots, \lambda_{2m}), \quad v = N^{-1/2}\sum_j e_{j+1}, \quad w = \sum_j c_je_{j+1},$$

$$\tau = vv^*, \quad L = e_0w^* + we_0^*, \quad \sigma_{\eta_0} = (1-\eta_0)\tau + \eta_0 I_{r_0}/r_0, \quad Z_0 = \operatorname{Tr} W.$$

The matrices $W, L, \tau, \sigma_{\eta_0}$ are rational even though v contains a square root, and $\|L\| \le \Lambda_0$. Set

$$\alpha = \rho \otimes \sigma_{\eta_0}, \qquad \beta(t) = (\rho \otimes W + tH \otimes L)/Z_0. \tag{74}$$

Both have trace one, since $\operatorname{Tr} L = 0$. The numerator is faithful. The relative tangent bound gives

$$\|(\rho \otimes W)^{-1/2}(H \otimes L)(\rho \otimes W)^{-1/2}\| \le h\Lambda_0 \le q_*.$$

Thus $\beta(t) > 0$ and its raw denominator is at least $(\rho \otimes W)/2$ when $|t| \le 1/(2q_*)$.

Uniform filter error. The implemented filter is $F(s) = F_m(s)/\sqrt{N}$, with mass $J_m = \int_0^\infty |F(s)|^2 ds = I_m/N$. For the beta-prime density proportional to f_m^2, direct integration gives

$$\mathbb{E}(T-1)^2 = \frac{2(m+1)}{m(2m-1)} \le \frac{2}{m} \quad (m \ge 2).$$

The multiplicative filter bounds in Lemma 3.2, including normalization, therefore imply for $\nu_m(ds) = F_m(s)^2 ds/I_m$ that

$$\mathbb{E}_{\nu_m}|s-1| \le e^{4/m+2/m^2}\sqrt{2/m} \le 3\sqrt{2/m} \quad (m \ge 6). \tag{75}$$

For eigenvalues a, b of either one output state,

$$\frac{|(b+as)^{-1} - (a+b)^{-1}|}{(a+b)^{-1}} = \frac{a|s-1|}{b+as} \le \kappa|s-1|.$$

Also if $K = A^{1/2}TA^{1/2}$ with $\|T\| \le h$, then

$$g_A(K) \le \operatorname{Tr}(KA^{-1}K) = \operatorname{Tr}(AT^2) \le h^2. \tag{76}$$

The first inequality follows from $2/(a+b) \le (1/a + 1/b)/2$ and Hermiticity of K. Let h_Φ be the pure-leaf-numerator entropy Hessian for channel Φ. The exact formula (12) and (75)–(76) give

$$|h_\Phi/J_m - g_{A_\Phi}(K_\Phi)| \le 3\kappa h^2\sqrt{2/m}.$$

By (72), the two-channel error is at most $d_0/2$. Hence $h_{\mathcal{Q}} - h_{\mathcal{P}} \ge J_m d_0/2$. For $s \in [1,2]$ all $\lambda_j \le 3$, so $F_m(s)^2 \ge 5^{-(4m+2)}$ and $J_m \ge j_*$. No numerical integration of the large partial-fraction coefficients is needed.

Dimension independent logarithm derivatives. For either output and any auxiliary density matrix σ, the variable part of its entropy is

$$-\operatorname{Tr}[(A\otimes\sigma)\log(B+tY)], \qquad B = A\otimes W, \quad Y = K\otimes L.$$

If $\mu = \lambda_{\min}(A)$, then $\|Y\| \le h\lambda_{\max}(A)\Lambda_0 \le q_*\mu$. For $|t| \le 1/(2q_*)$, $B + tY \ge \mu I/2$. Resolvent differentiation gives

$$\frac{d^n}{dt^n}\log(B+tY) = (-1)^{n+1}n!\int_0^\infty R_t(u)[YR_t(u)]^n\,du, \quad R_t(u) = (B+tY+uI)^{-1}.$$

Taking norms and integrating yields

$$\|\frac{d^n}{dt^n}\log(B+tY)\| \le (n-1)!\left(\frac{\|Y\|}{\lambda_{\min}(B+tY)}\right)^n.$$

Since $\operatorname{Tr}(A\otimes\sigma) = 1$, each channel's absolute second derivative at zero is at most q_*^2, and its absolute third derivative throughout the interval is at most $16q_*^3$.

Faithful numerator and a finite perturbation. Let $G_\sigma(t)$ be the target-minus-source entropy difference with numerator $\rho\otimes\sigma$ and denominator $\beta(t)$. Saturation gives $G_\sigma(0) = 0$. For $\sigma = \tau, I_{r_0}/r_0, \sigma_{\eta_0}$, conjugating the centre coordinate by -1 leaves the numerator fixed and sends L to $-L$. Thus G_σ is even and $G'_\sigma(0) = 0$. The second denominator derivative is linear in σ. Hence

$$G''_\tau(0) \ge j_* d_0/2, \qquad |G''_{I_{r_0}/r_0}(0)| \le 2q_*^2,$$

and the choice of η_0 implies

$$G''_{\sigma_{\eta_0}}(0) \ge \tfrac{3}{4}(j_* d_0/2) - 2\eta_0 q_*^2 \ge j_* d_0/4.$$

The two-channel third derivative has absolute value at most $32q_*^3$. Taylor's theorem now gives

$$G_{\sigma_{\eta_0}}(t) \ge (j_* d_0/8)t^2 - (16/3)q_*^3|t|^3.$$

At $t = t_0$, the cubic term is at most $(j_* d_0/16)t_0^2$, so the gap is at least ε_0. Finally $\sigma_{\eta_0} \le I \le W$ and the denominator lower bound imply $\beta(t) \ge \alpha/(2Z_0)$. This inequality survives either channel. Operator monotonicity of logarithm gives each witness divergence in $[0, \log(2Z_0)]$. Thus the construction involves finite faithful divergences and no infinity-minus-infinity expression. □

C Uniform bounds for POVM completion

We justify the estimate used in (61) and make the choice of the signal parameter effective. All norms below use the common reference marginal W, so no smallest eigenvalue of a reference state enters a constant.

Choose rational lower bounds $D_i \geq d_i I$, $d_i > 0$, and certified rational upper bounds in the following definitions:

$$\begin{aligned} \kappa &\geq \max_{i,s=E,F} \|O_i^s\|_{\mathrm{op}}/d_i, \\ B_z &\geq \sum_i d_i^{-1}(\|O_i^E\|_{\mathrm{HS}}^2 + \|O_i^F\|_{\mathrm{HS}}^2), \\ B_u &\geq 2\sum_i d_i^{-1}\|D_i\|_{\mathrm{HS}}^2. \end{aligned} \tag{77}$$

We may take all positive; a zero signal case is trivial. Since $R_i \geq d_i W$ and $-\|O\|_{\mathrm{op}} W \leq V \leq \|O\|_{\mathrm{op}} W$, one has $\|T\| \leq \kappa$ uniformly. Coefficient-frame Cauchy–Schwarz and inverse Jordan ordering imply

$$\sum_{i,s} \|z_i^s\|^2 \leq B_z |X_{\mathcal{C}}|_W^2, \qquad \sum_{i,s} \|u_i\|^2 \leq B_u |Y|_W^2.$$

For $\delta\kappa \leq 1/2$,

$$0 \leq (I - \delta^2 T^2)^{-1} - I \leq \tfrac{4}{3}\delta^2\kappa^2 I.$$

Let $a = |X_{\mathcal{C}}|_W$, $b = |Y|_W$, and $\gamma = \gamma_{\mathrm{cross}}$. The gap (58) applies to the pinching of R; its R_i and W are exactly unchanged by that pinching. Thus the leading z-square source-minus-target difference is at least γa^2. Expanding (60), bounding the mixed terms by Cauchy–Schwarz, and using the last resolvent bound gives

$$\begin{aligned} \delta^{-2}\Delta_{\mathrm{signal}} \geq \gamma a^2 - 2\kappa\sqrt{B_z B_u}\, ab - \kappa^2 B_u b^2 \\ - \tfrac{8}{3}\delta^2\kappa^2(B_z a^2 + \kappa^2 B_u b^2). \end{aligned} \tag{78}$$

This includes all physical coherences through V and T. If

$$\delta^2 \leq \frac{3\gamma}{16\kappa^2 B_z}, \qquad \delta\kappa \leq \tfrac{1}{2}, \tag{79}$$

then the resolvent cross loss is at most $\gamma a^2/2$. Young's inequality absorbs the mixed term using another $\gamma a^2/4$ and $4\kappa^2 B_z B_u b^2/\gamma$. The diagonal resolvent loss is at most $(2/3)\kappa^2 B_u b^2$. Hence

$$\Delta_{\mathrm{signal}} \geq \delta^2 \left[\frac{\gamma}{4} a^2 - C_0 b^2\right], \qquad C_0 = \frac{5}{3}\kappa^2 B_u + \frac{4\kappa^2 B_z B_u}{\gamma}. \tag{80}$$

The final POVMs give equal weight $1/2$ to signal and padding. Equations (59) and (80) prove nonnegative total Fisher difference whenever, in addition,

$$\delta^2 \leq c/C_0. \tag{81}$$

Choose a positive dyadic rational δ satisfying (79)–(81). The same condition $\delta\kappa \leq 1/2$ ensures $D_i \pm \delta O_i^s \geq (d_i/2)I$, so all final effects are strictly positive.

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